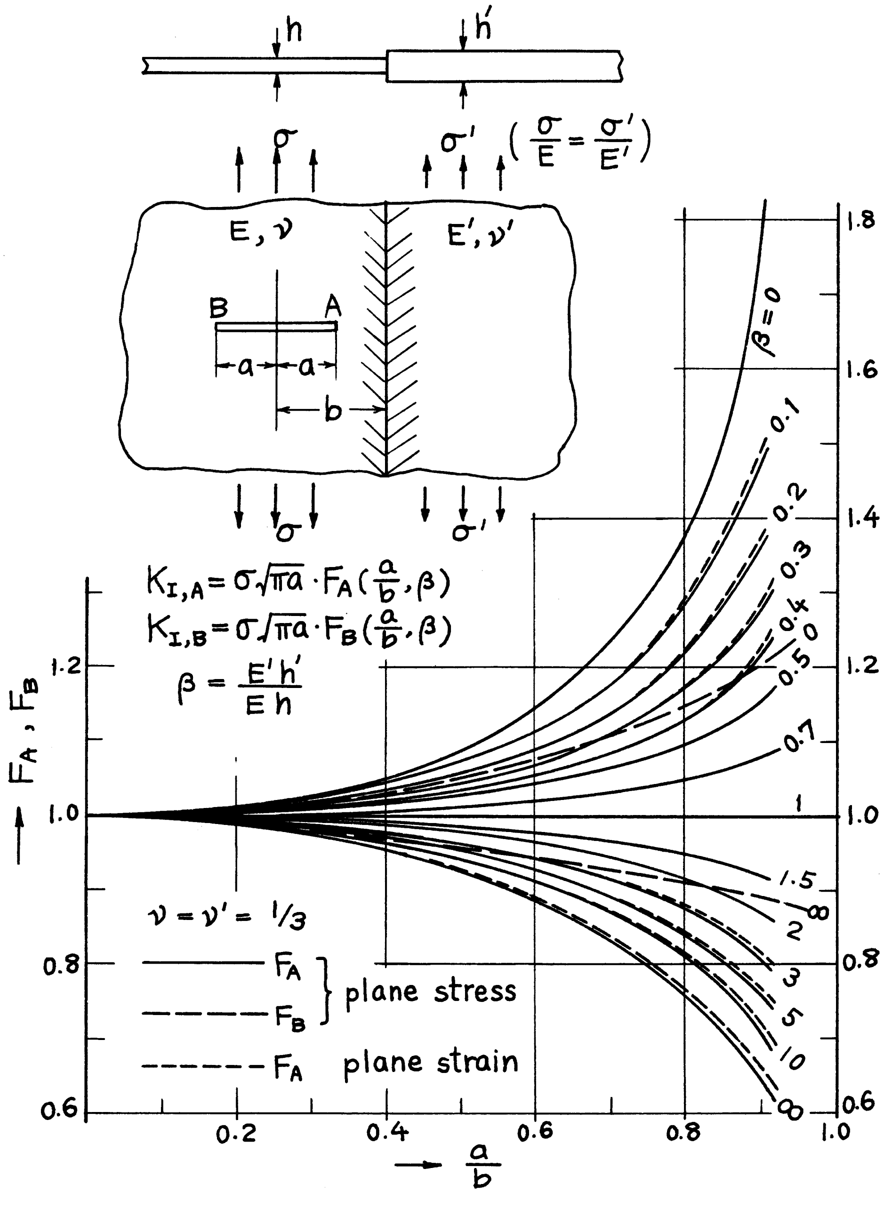
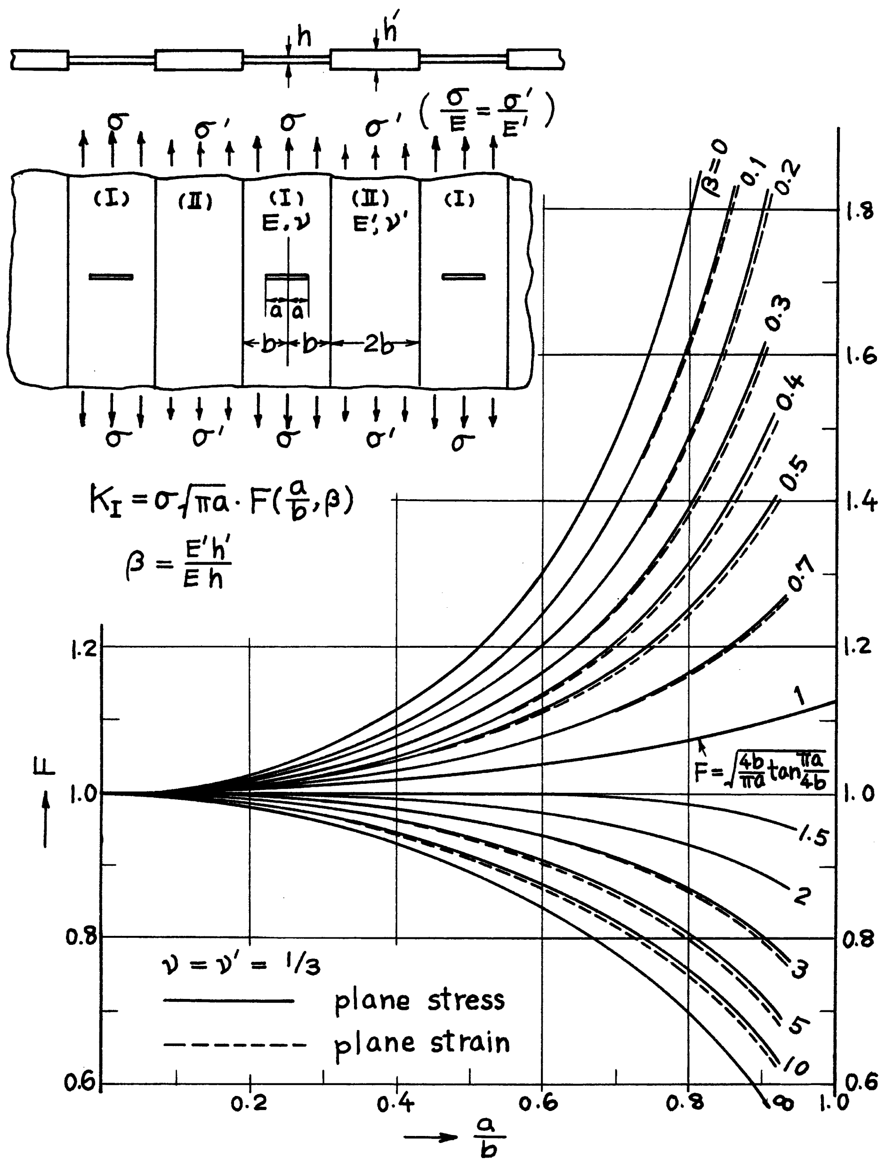


Method: Expansions of Complex Stress Potentials
Accuracy: Curves were drawn based on the results having 0.1% accuracy.
Reference: **Isida 1970a**



Method: Expansions of Complex Stress Potentials

Accuracy: Curves were drawn based on results with 0.1% accuracy.

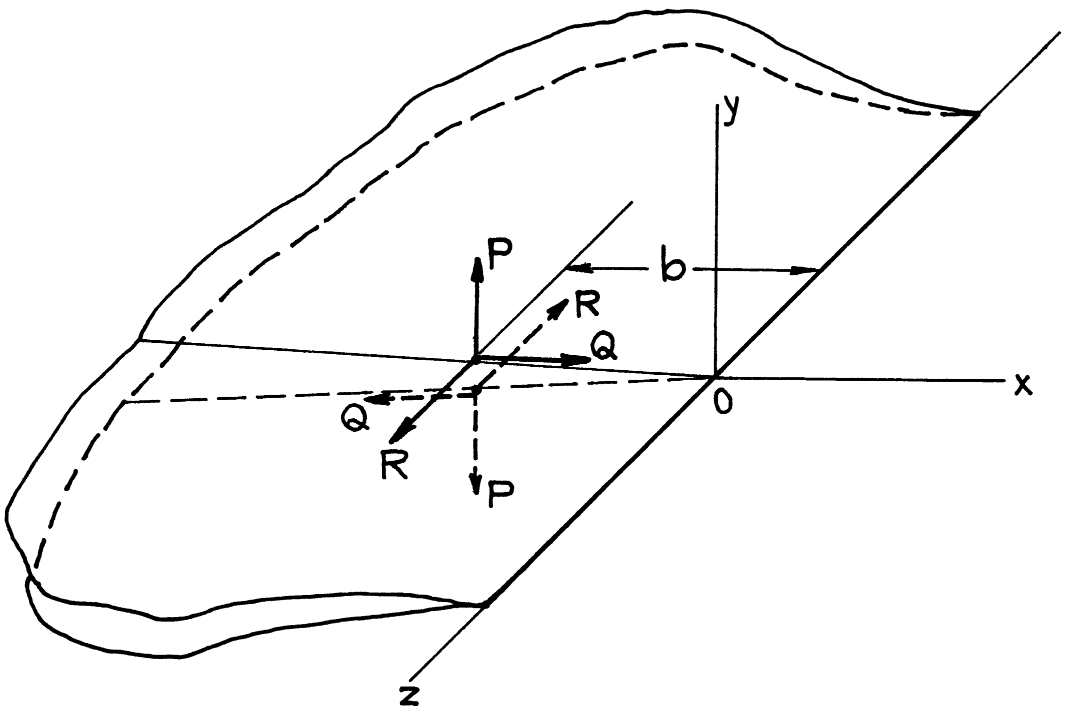
Reference: **Isida 1970a**

PART

IV

THREE DIMENSIONAL CRACKED CONFIGURATIONS

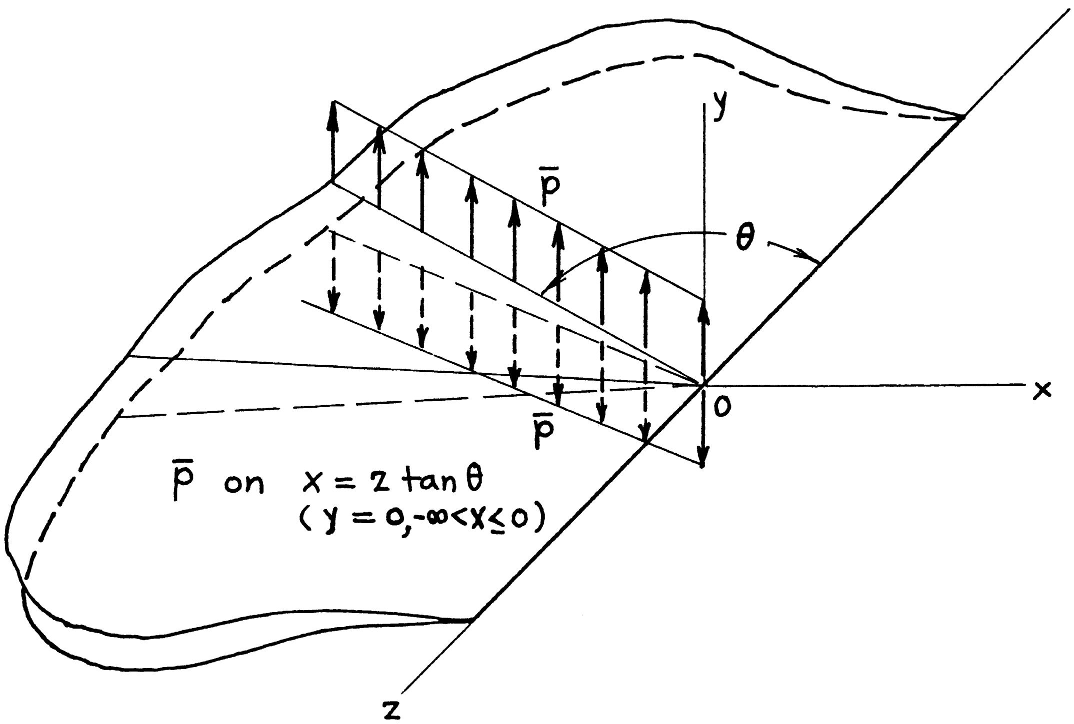
- A Semi-Infinite Crack in an Infinite Body
- An Embedded Circular Crack in an Infinite Body
- An External Circular Crack (A Circular Net Section) or a Circular Ring (An Annular) Crack in an Infinite Body
- An Elliptical Crack or Net Section and a Parabolic Crack in an Infinite Body
- An External Circular Crack in a Round Bar
- An Internal Circular Crack in a Round Bar
- An Internal Circumferential Crack in a Thick-Walled Cylinder
- An External Circumferential Crack in a Thick-Walled Cylinder
- A Half-Circular Surface Crack in a Semi-Infinite Body
- A Quarter Circular Corner Crack in a Quarter-Infinite Body



$$K_I = \frac{\sqrt{2}P}{(\pi b)^{3/2}} \cdot \frac{1}{1 + \left(z/b\right)^2}$$

$$\begin{Bmatrix} K_{II} \\ K_{III} \end{Bmatrix} = \frac{\sqrt{2}}{(\pi b)^{3/2}} \cdot \begin{Bmatrix} Q \\ R \end{Bmatrix} \frac{1}{1 + \left(z/b\right)^2} \left[1 \begin{Bmatrix} + \\ - \end{Bmatrix} \frac{2\nu}{2-\nu} \cdot \frac{1 - \left(z/b\right)^2}{1 + \left(z/b\right)^2} \right] + \frac{4\sqrt{2}}{(\pi b)^{3/2}} \begin{Bmatrix} R \\ Q \end{Bmatrix} \cdot \frac{\nu}{2-\nu} \cdot \frac{z/b}{\left[1 + \left(z/b\right)^2\right]^2}$$

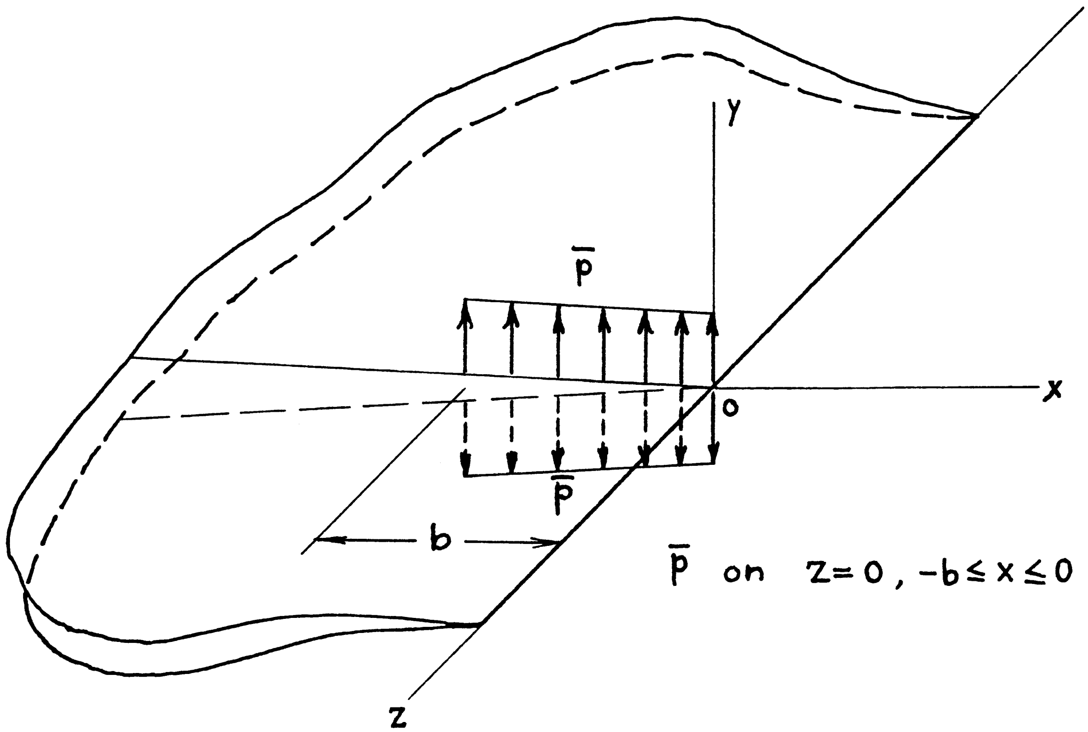
Method: Papkovich-Neuber Potentials
Accuracy: Exact
References: **Uflyand 1965; Sih 1968; Kassir 1973**



$$K_I(z;\ \theta) = \frac{\overline{p}}{\sqrt{\pi |z|}} \begin{cases} \sqrt{\tan\frac{\theta}{2}} & z > 0 \\ \sqrt{\cot\frac{\theta}{2}} & z < 0 \end{cases}$$

$$K_{II} = K_{III} = 0$$

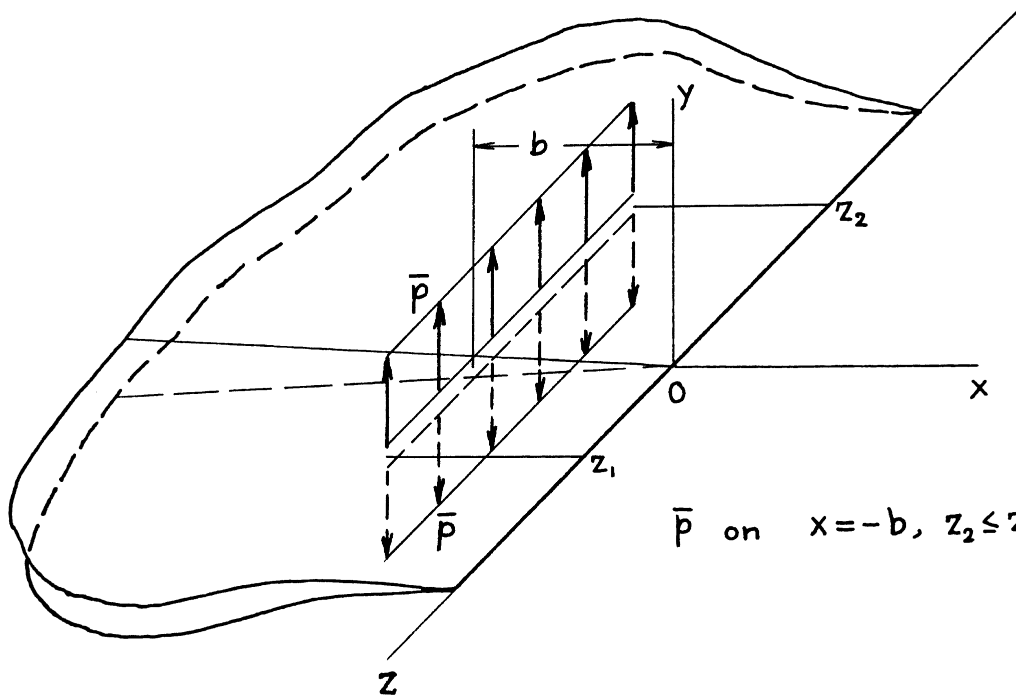
Method: Integration of **page 23.1** or a Limiting Case of **page 24.11**
Accuracy: Exact
Reference: **Tada 1985**



$$K_I(z\,;b) = \frac{\bar{p}}{\pi^{3/2}\sqrt{|z|}} \cdot \begin{cases} \pi - \sin^{-1}\sqrt{\dfrac{2\frac{|z|}{b}}{1+\left(\frac{z}{b}\right)^2}} - \sinh^{-1}\sqrt{\dfrac{2\frac{|z|}{b}}{1+\left(\frac{z}{b}\right)^2}} & \dfrac{|z|}{b} \le 1 \\ \sin^{-1}\sqrt{\dfrac{2\frac{|z|}{b}}{1+\left(\frac{z}{b}\right)^2}} - \sinh^{-1}\sqrt{\dfrac{2\frac{|z|}{b}}{1+\left(\frac{z}{b}\right)^2}} & \dfrac{|z|}{b} > 1 \end{cases}$$

$$K_{II} = K_{III} = 0$$

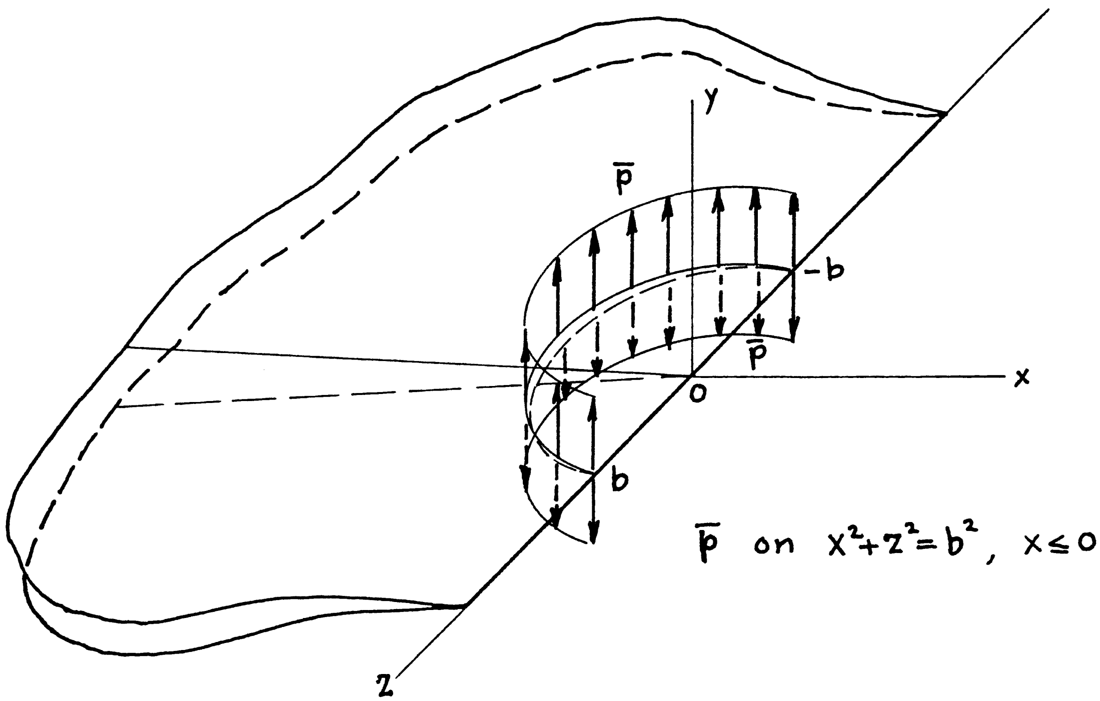
Method: Integration of **page 23.1**
Accuracy: Exact
Reference: **Tada 1985**



$$K_I(z) = \frac{\sqrt{2}\,\bar{p}}{\pi^{3/2}\sqrt{b}}\left[\tan^{-1}\frac{z-z_1}{b} - \tan^{-1}\frac{z-z_2}{b}\right]$$

$$K_{II} = K_{III} = 0$$

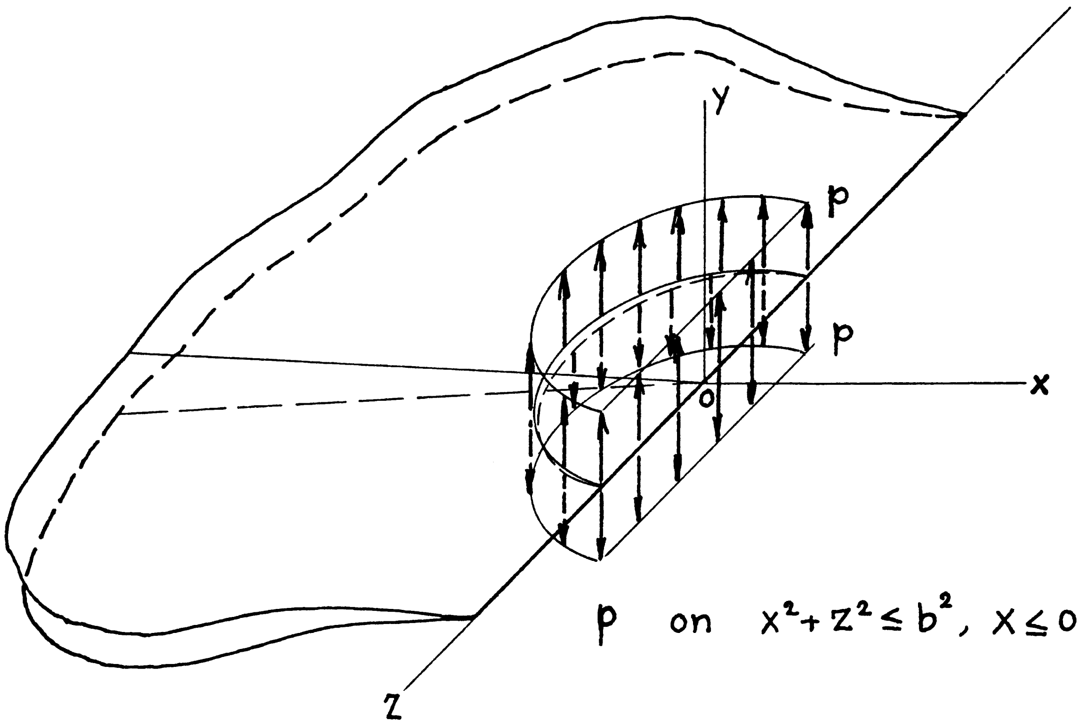
Method: Integration of **page 23.1** or a Limiting Case of **page 24.4**
Accuracy: Exact
Reference: **Tada 1985**



$$K_I(z) = \frac{\sqrt{2}\,\bar{p}}{\pi\sqrt{b}} \sum_{n=0}^{\infty} \frac{\Gamma\left(n+\frac{3}{4}\right)}{\Gamma\left(n+\frac{5}{4}\right)} \left(\frac{b}{z}\right)^{2n+2}$$

$$K_{II} = K_{III} = 0$$

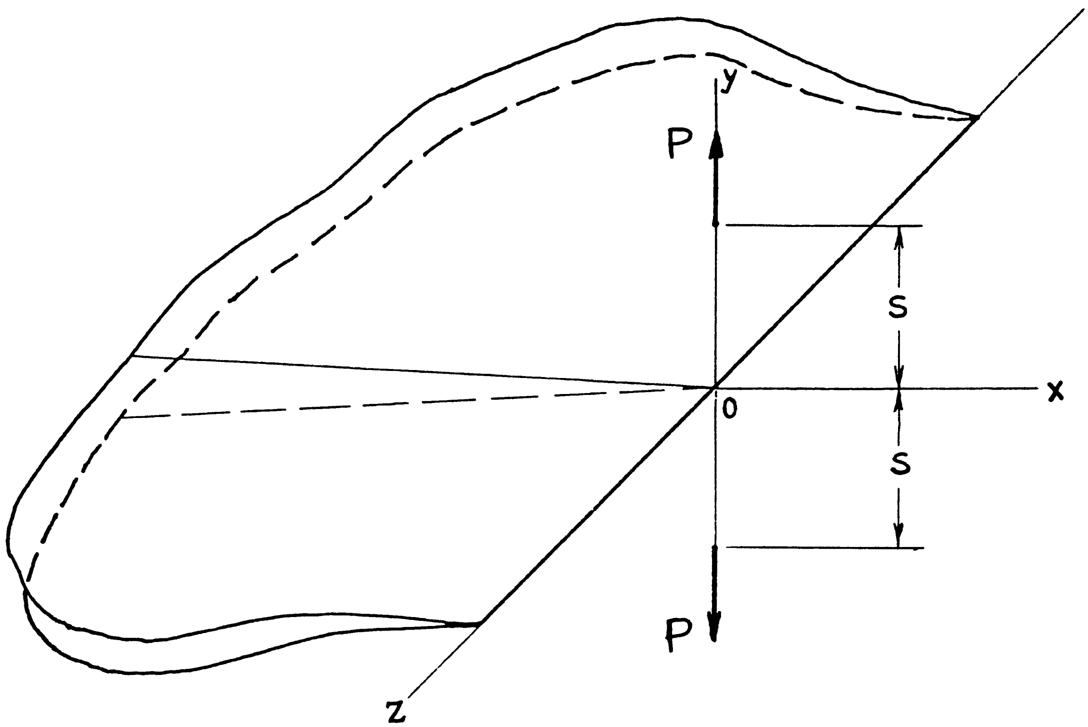
Method: Integration of **page 23.1**
Accuracy: Exact
Reference: **Tada 1985**



$$K_I(z) = \frac{P\sqrt{b}}{\sqrt{2}\pi} \sum_{n=0}^{\infty} \frac{\Gamma\left(n+\frac{3}{4}\right)}{\left(n+\frac{5}{4}\right)\Gamma\left(n+\frac{5}{4}\right)} \left(\frac{b}{z}\right)^{2n+2}$$

$$K_{II} = K_{III} = 0$$

Method: Integration of **page 23.1 (or 23.5)**
Accuracy: Exact
Reference: **Tada 1985**



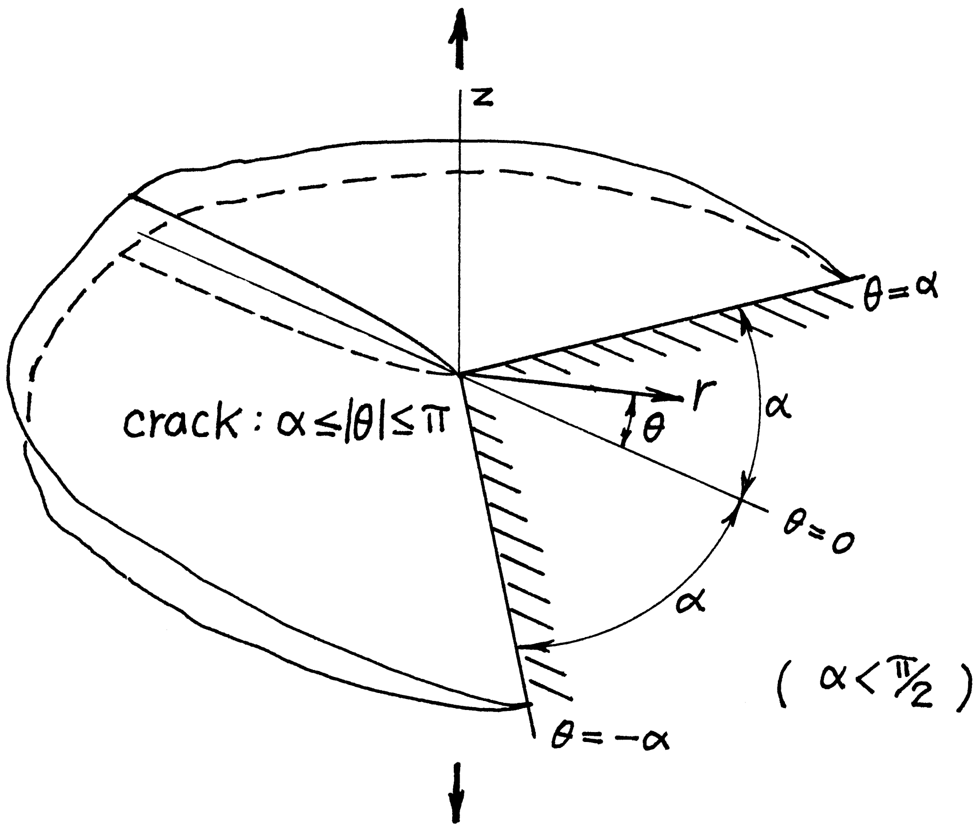
$$K_I(z) = \frac{P}{\pi^{3/2}}\left(1 - \alpha s\,\frac{\partial}{\partial s}\right)\frac{\sqrt{s}}{s^2 + z^2}$$

$$= \frac{P}{(\pi s)^{3/2}}\frac{1}{1+\left(\frac{z}{s}\right)^2}\left[1 - \frac{\alpha}{2}\left\{1 - \frac{4}{1+\left(\frac{z}{s}\right)^2}\right\}\right]$$

$$\text{where}\quad \alpha = \frac{1}{2(1-\nu)}$$

$$K_{II} = K_{III} = 0$$

Method: Integration of **page 23.1 (or 23.5)**
Accuracy: Exact
References: **Tada 1985; also Kassir 1975**

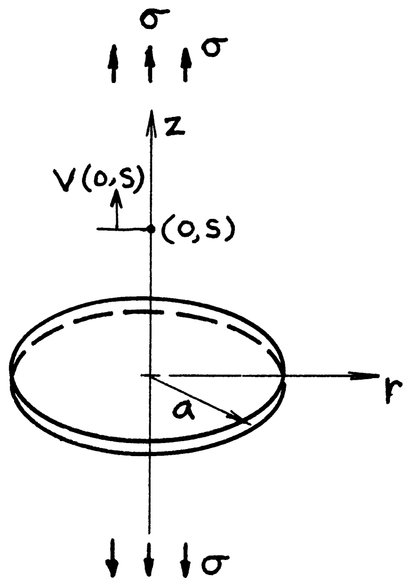


$$K_I = \frac{p}{\sqrt{\pi r}\ \sin\ \alpha}$$

$$K_{II} = K_{III} = 0$$

where P = Total Line Load on Arc: r = const., $-\alpha \leq \theta \leq \alpha$
= Constant for any r

Method: Neuber-Papkovich Potential
Accuracy: Exact
References: **Galin 1953; Tada 1974**



$$K_I = \frac{2}{\pi}\sigma\sqrt{\pi a} \qquad (K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{16\left(1-\nu^2\right)}{3E}\sigma a^3$$

Crack Opening Shape:

$$2v\underset{r\le a}{(r,0)} = \frac{8\left(1-\nu^2\right)}{\pi E}\sigma\sqrt{a^2 - r^2}$$

Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{8\left(1-\nu^2\right)}{\pi E}\sigma a$$

Additional Displacement at $(0, s)$ due to Crack:

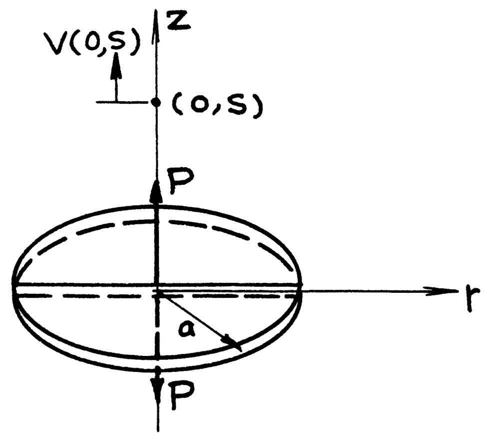
$$v(0,s) = \frac{4\left(1-\nu^2\right)}{\pi E}\sigma a\left\{1 - (1-\alpha)\frac{s}{a}\tan^{-1}\frac{a}{s} - \alpha\frac{s^2}{s^2 + a^2}\right\}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Methods: Integral Transform, Integration of **page 24.5 or 24.11**, Paris' Equation (see **Appendix B**), Reciprocity (see **page 24.7**)
Accuracy: Exact
Reference: **Tada 1985**

NOTE: $V(0,s)$ Is the displacement at $(0,s)$ when uniform pressure σ is applied on crack surfaces.



$$K_I = \frac{P}{(\pi a)^{3/2}} \qquad (K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{8\left(1-\nu^2\right)}{\pi E} P\, a$$

Crack Opening Shape:

$$\underset{r \le a}{2v\ (r, 0)} = \frac{4\left(1-\nu^2\right)}{\pi^2 E} P \frac{1}{r} \cos^{-1} \frac{r}{a}$$

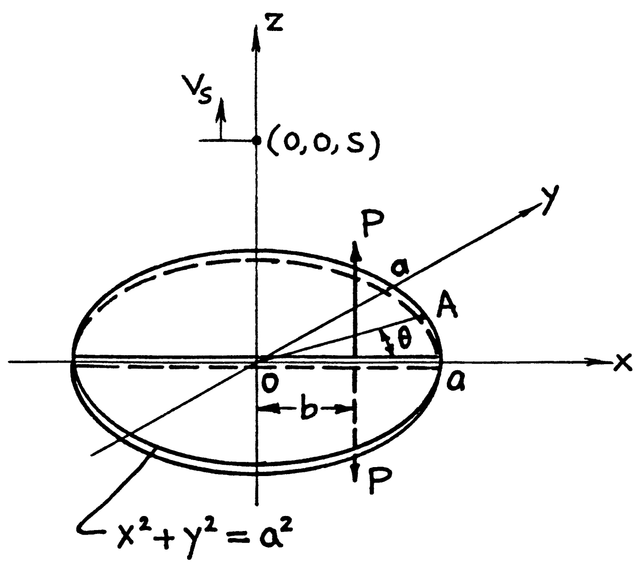
Displacement at $(0, s)$:

$$v(0, s) = \frac{2\left(1-\nu^2\right)}{\pi^2 E} P \frac{1}{s} \left\{ (1+\alpha) \tan^{-1} \frac{a}{s} + \alpha \frac{as}{s^2 + a^2} \right\}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Methods: Special case of **page 24.3**, Reciprocity (see **page 24.3 or 24.7**)
Accuracy: Exact
References: **Galin 1953; Tada 1973, 1985**



$$K_{IA}=K_I(\theta)=\frac{P}{\pi\sqrt{\pi a}}\cdot\frac{\sqrt{a^2-b^2}}{a^2+b^2-2ab\cos\theta}$$

$$(K_{II}=K_{III}=0)$$

Volume of Crack:

$$V=\frac{8\left(1-\nu^2\right)}{\pi E}P\sqrt{a^2-b^2}$$

Crack Opening at Center:

$$\delta_0=2v(0,0,0)=\frac{4\left(1-\nu^2\right)}{\pi^2 E}P\frac{1}{b}\cos^{-1}\frac{b}{a}$$

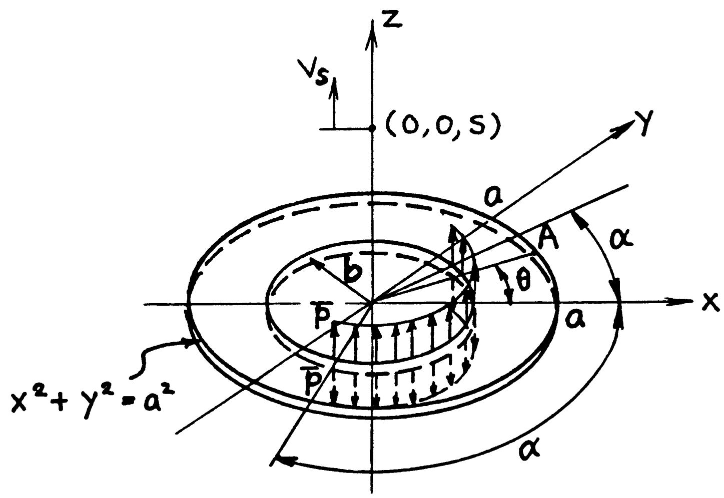
Vertical Displacement at $(0,0,s)$:

$$v_s=v(0,0,s)=\frac{2\left(1-\nu^2\right)}{\pi^2 E}P\left(1-\alpha s\frac{\partial}{\partial s}\right)\frac{1}{\sqrt{s^2+b^2}}\tan^{-1}\sqrt{\frac{a^2-b^2}{s^2+b^2}}$$

where

$$\alpha=\frac{1}{2(1-\nu)}$$

Methods: Neuber-Papkovich Potentials, Reciprocity (see **pages 24.1, 24.2, 24.7**)
Accuracy: Exact
References: **Galin 1953; Tada 1985**



$$K_{IA} = K_I(\theta) = \frac{2P}{\pi\sqrt{\pi a}} \frac{b}{\sqrt{a^2 - b^2}} \left\{ \tan^{-1}\left(\frac{a+b}{a-b}\tan\frac{\theta+\alpha}{2}\right) - \tan^{-1}\left(\frac{a+b}{a-b}\tan\frac{\theta-\alpha}{2}\right) \right\}$$

$$(K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{16\left(1-\nu^2\right)\alpha}{\pi E}\bar{p}b\sqrt{a^2 - b^2}$$

Crack Opening at Center:

$$\delta_0 = 2v(0,0,0) = \frac{8\left(1-\nu^2\right)\alpha}{\pi^2 E}\bar{p}\cos^{-1}\frac{b}{a}$$

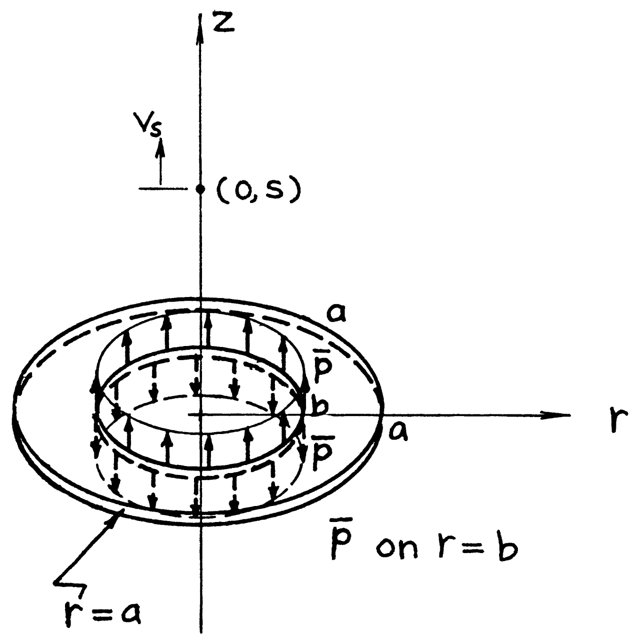
Vertical Displacement at $(0, 0, s)$:

$$v_s = v(0,0,s) = \frac{4\left(1-\nu^2\right)\alpha}{\pi^2 E}\bar{p}\left(1 - \alpha' s\frac{\partial}{\partial s}\right)\frac{b}{\sqrt{s^2 + b^2}}\tan^{-1}\sqrt{\frac{a^2 - b^2}{s^2 + b^2}}$$

where

$$\alpha' = \frac{1}{2(1-\nu)}$$

Method: Integration of **page 24.3**
Accuracy: Exact
References: **Tada 1973, 1985**



$$K_I = \frac{2\overline{p}}{\sqrt{\pi a}} \frac{b}{\sqrt{a^2 - b^2}} \qquad (K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{16\left(1-\nu^2\right)}{E} \overline{p}\, b \sqrt{a^2 - b^2}$$

Crack Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{8\left(1-\nu^2\right)}{\pi E} \overline{p} \cos^{-1} \frac{b}{a}$$

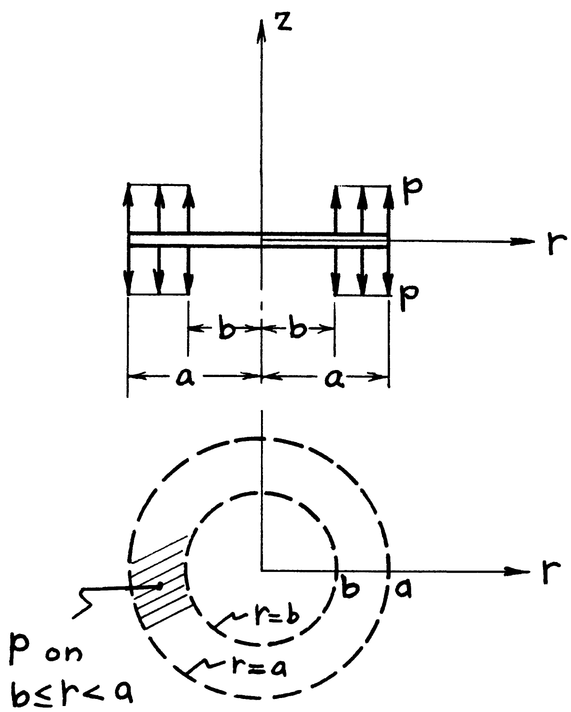
Displacement at $(0, s)$:

$$v_s = v(0,s) = \frac{4\left(1-\nu^2\right)}{\pi E} \overline{p} \left(1 - \alpha s \frac{\partial}{\partial s}\right) \frac{b}{\sqrt{s^2 + b^2}} \tan^{-1} \sqrt{\frac{a^2 - b^2}{s^2 + b^2}}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Methods: Integral Transform, Weight Function Method, Special Case of **page 24.4**
Accuracy: Exact
References: **Sneddon 1946, 1951; Barenblatt 1962; Bueckner 1972; Tada 1985**



$$K_I = \frac{2p}{\sqrt{\pi a}} \sqrt{a^2 - b^2} \qquad (K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{16\left(1 - \nu^2\right)}{3E} p a^3 \left\{1 - \left(\frac{b}{a}\right)^2\right\}^{3/2}$$

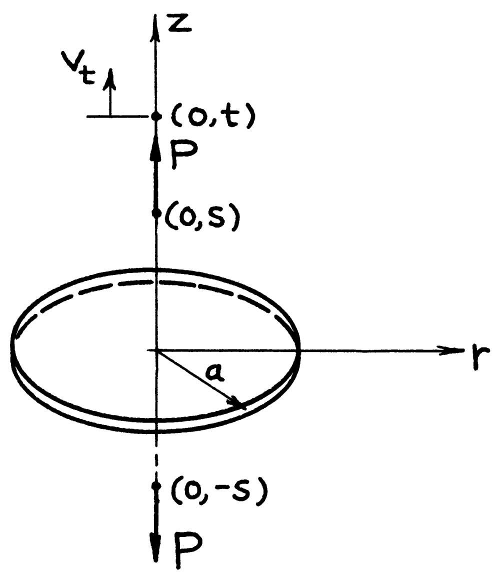
Crack Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{8\left(1 - \nu^2\right)}{\pi E} pa \left\{\sqrt{1 - \left(\frac{b}{a}\right)^2} - \frac{b}{a} \cos^{-1} \frac{b}{a}\right\}$$

Crack Opening at $r = b$:

$$\delta_b = 2v(b, 0) = \frac{8\left(1 - \nu^2\right)}{\pi E} pa \left(1 - \frac{b}{a}\right)$$

Methods: Integral Transform, Integration of **page 24.5**
Accuracy: Exact
References: **Sneddon 1951; Barenblatt 1962; Tada 1985**



$$K_I = \frac{P}{\pi^2}\sqrt{\pi a}\left(1 - \alpha s \frac{\partial}{\partial s}\right)\frac{1}{a^2+s^2}$$

$$or \quad = \frac{P}{(\pi a)^{3/2}}\frac{a^2}{a^2+s^2}\left\{1 + \alpha\frac{2s^2}{a^2+s^2}\right\}$$

$$(K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{8\left(1-\nu^2\right)}{\pi E}Pa\left\{1 - (1-\alpha)\frac{s}{a}\tan^{-1}\frac{a}{s} - \alpha\frac{s^2}{a^2+s^2}\right\}$$

Crack Opening Shape:

$$\underset{r \le a}{2v(r,0)} = \frac{4\left(1-\nu^2\right)}{\pi^2 E}P\left(1 - \alpha s\frac{\partial}{\partial s}\right)\frac{1}{\sqrt{s^2+r^2}}\tan^{-1}\sqrt{\frac{a^2-r^2}{s^2+r^2}}$$

Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{4\left(1-\nu^2\right)}{\pi^2 E}P\cdot\frac{1}{s}\left\{(1+\alpha)\tan^{-1}\frac{a}{s} + \alpha\frac{as}{a^2+s^2}\right\}$$

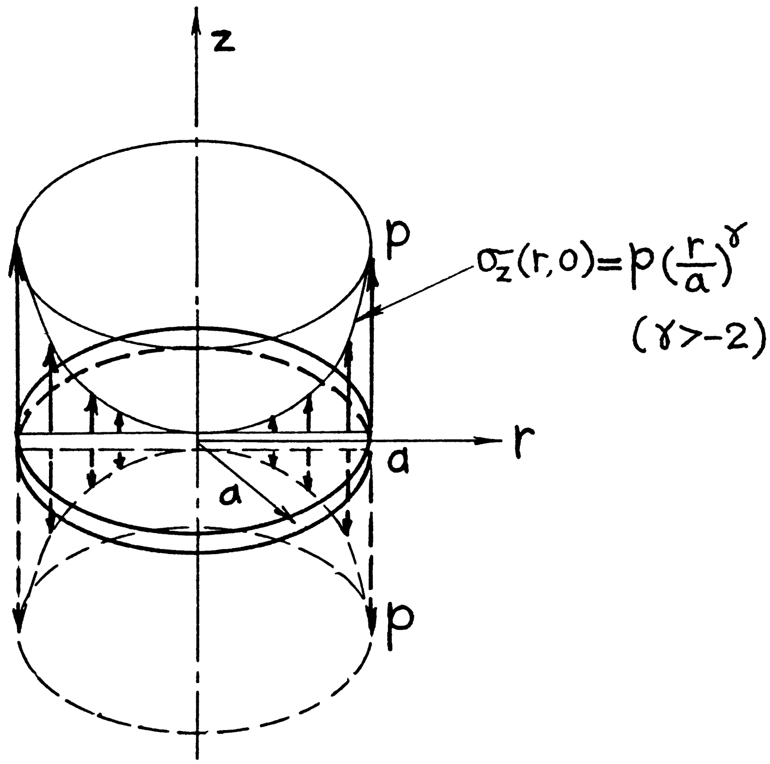
Displacement at $(0,t)$:

$$v_t = v(0,t) = \frac{4\left(1-\nu^2\right)}{\pi^2 E}P\left(1 - \alpha s\frac{\partial}{\partial s}\right)\left(1 - \alpha t\frac{\partial}{\partial t}\right)\left\{\frac{t}{t^2-s^2}\tan^{-1}\frac{a}{t} + \frac{s}{s^2-t^2}\tan^{-1}\frac{a}{s}\right\}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Methods: Integration of **page 24.5**, Paris' Equation (see **Appendix B**)
Accuracy: Exact
References: **Barenblatt 1962; Tada 1985**



$$K_I = P\sqrt{a} \cdot \frac{\Gamma\left(\frac{\gamma}{2}+1\right)}{\Gamma\left(\frac{\gamma}{2}+\frac{3}{2}\right)}$$

$$(K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{4\left(1-\nu^2\right)}{E} pa^3 \sqrt{\pi} \cdot \frac{\Gamma\left(\frac{\gamma}{2}+1\right)}{\Gamma\left(\frac{\gamma}{2}+\frac{5}{2}\right)}$$

Crack Opening at Center:

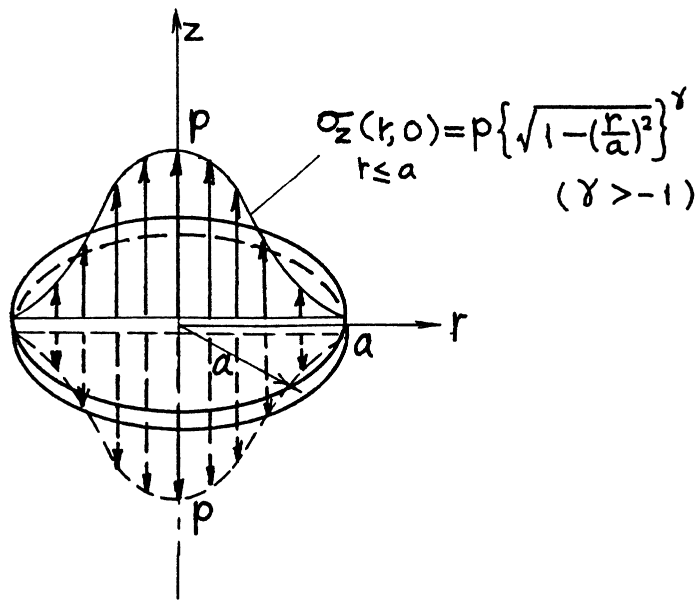
$$\delta_0 = 2v(0,0) = \frac{4\left(1-\nu^2\right)}{E} pa \frac{1}{\sqrt{\pi}} \cdot \frac{\Gamma\left(\frac{\gamma}{2}+1\right)}{(\gamma+1)\Gamma\left(\frac{\gamma}{2}+\frac{3}{2}\right)}$$

where

$$\Gamma(\gamma) = \text{Gamma Function (see } \textbf{Appendix M)}$$

Method: Integration of **page 24.5**, Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1974, 1985**

NOTE: For special case of $\gamma = 0$, see **page 24.1**.



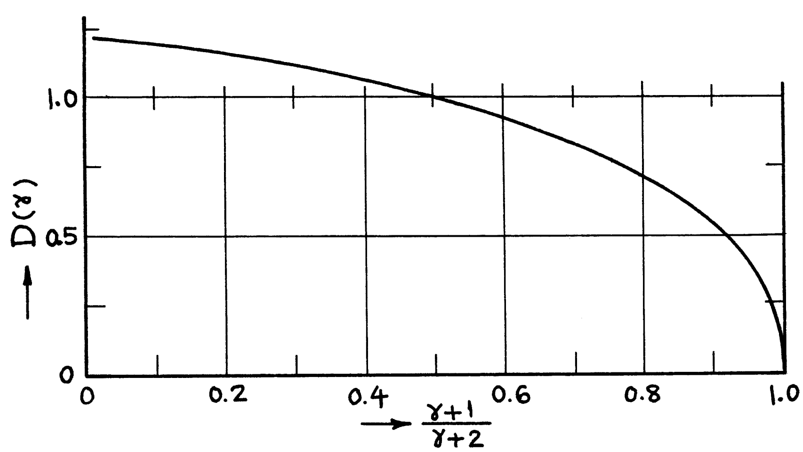
$$K_I = \frac{2}{\pi} P\sqrt{\pi a}\left(\frac{1}{\gamma+1}\right) \qquad (K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{16\left(1-\nu^2\right)}{3E} pa^3\left(\frac{3}{\gamma+3}\right)$$

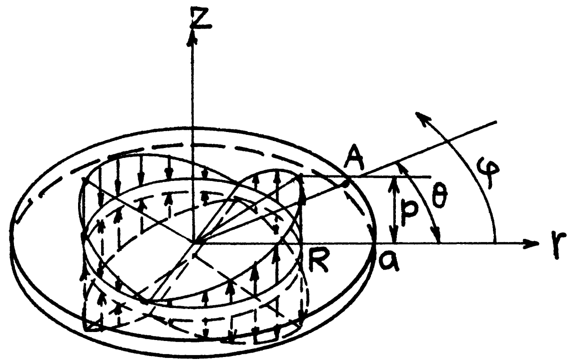
Crack Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{8\left(1-\nu^2\right)}{\pi E} pa \cdot D(\gamma)$$



Method: Integration of **page 24.5**
Accuracy: K_I, V Exact; $D(\gamma)$ curve is based on accurate numerical values.
References: **Tada 1974, 1985**

NOTE: For special case of $\gamma = 0$, see **page 24.1**.

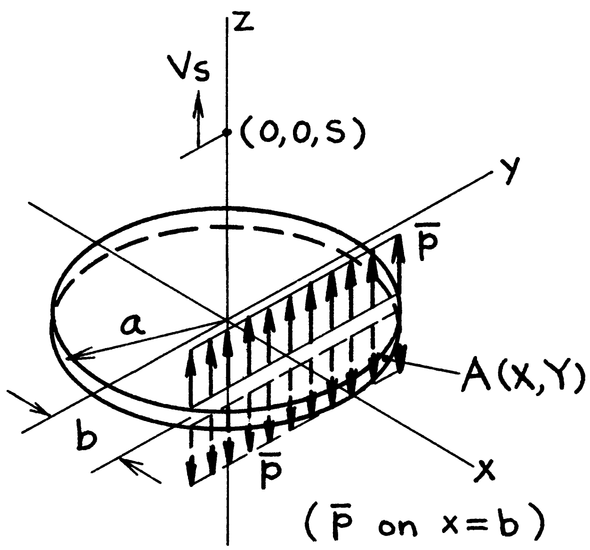


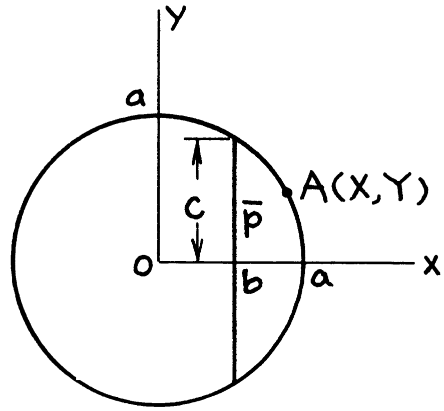
$$\sigma_z(r = R, \varphi) = p \cdot \cos\varphi$$

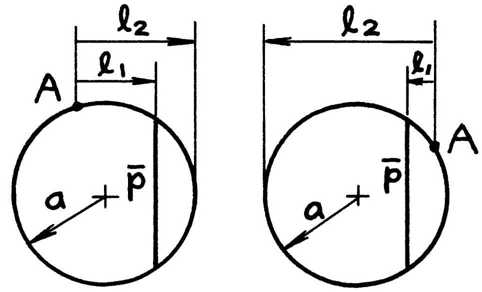
$$K_{IA} = \frac{2P}{a\sqrt{\pi a}} \frac{R^2}{\sqrt{a^2 - R^2}} \cos\theta$$

$$K_{IIA} = K_{IIIA} = 0$$

Method: Integration of **page 24.3**
Accuracy: Exact
Reference: **Tada 1973**







$$K_{IA} = \frac{\overline{p}}{\sqrt{\pi a}}\left(\sqrt{\frac{\ell_2}{\ell_1}} - 1\right)$$

$$= \frac{\overline{p}}{\sqrt{\pi a}}\left(\sqrt{\frac{a}{|X-b|} + \frac{X}{X-b}} - 1\right)$$

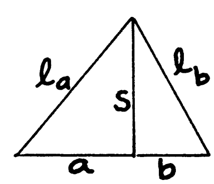
$$or \quad K_{IA} = \begin{cases} \dfrac{\overline{p}}{\sqrt{\pi a}}\left(\sqrt{\frac{a+X}{X-b}} - 1\right) & X > b \\ \dfrac{\overline{p}}{\sqrt{\pi a}}\left(\sqrt{\frac{a-X}{b-X}} - 1\right) & X < b \end{cases}$$

Volume of Crack:

$$V = \frac{4\left(1-\nu^2\right)}{E}\,\overline{p}\left(a^2 - b^2\right) = \frac{4\left(1-\nu^2\right)}{E}\,\overline{p}\,c^2$$

Crack Opening at Center:

$$\delta_0 = 2v(0,0,0) = \frac{4\left(1-\nu^2\right)}{\pi E}\,\overline{p}\,\ell n\frac{a}{|b|}$$



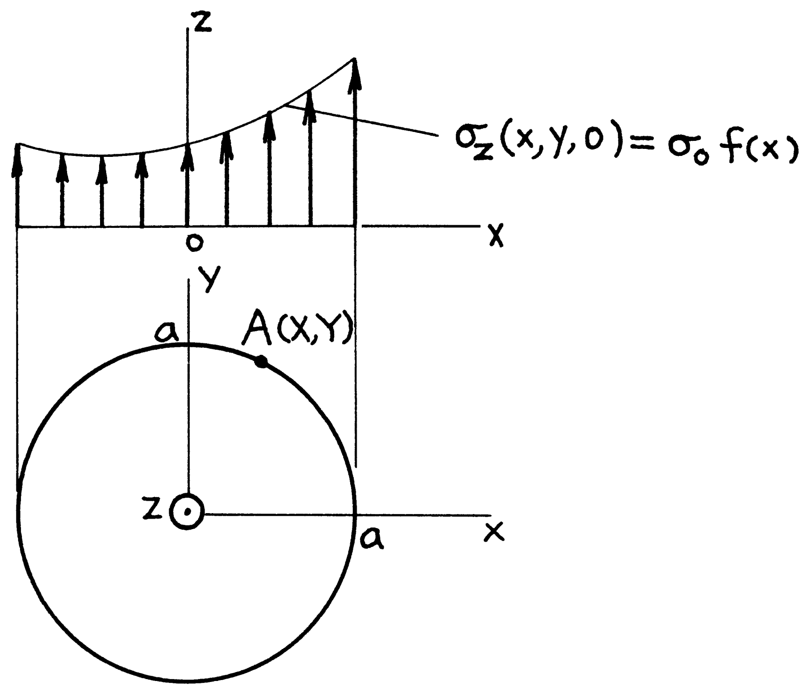
Vertical Displacement at $(0,0,s)$:

$$v_s = v(0,0,s) = \frac{2\left(1-\nu^2\right)}{\pi E}\,\overline{p}\left(1 - \alpha s\frac{\partial}{\partial s}\right)\ell n\frac{\ell_a}{\ell_b}$$

$$= \frac{2\left(1-\nu^2\right)}{\pi E}\,\overline{p}\left\{\ell n\sqrt{\frac{a^2+s^2}{b^2+s^2}} - \alpha\left(\frac{s^2}{a^2+s^2} - \frac{s^2}{b^2+s^2}\right)\right\}$$

$$\alpha = \frac{1}{2(1-\nu)}$$

Methods: Integration of **page 24.2**, or V, δ_0, v_s Paris' Equation (see **Appendix B**)
Accuracy: Exact
References: **Tada 1975, 1985**



$$K_{IA} = \frac{\sigma_0}{\sqrt{\pi a}} \left\{ \sqrt{a+X} \int_{-a}^{X} \frac{f(x)}{\sqrt{X-x}} \, dx + \sqrt{a-X} \int_{X}^{a} \frac{f(x)}{\sqrt{x-X}} \, dx - \int_{-a}^{a} f(x) dx \right\}$$

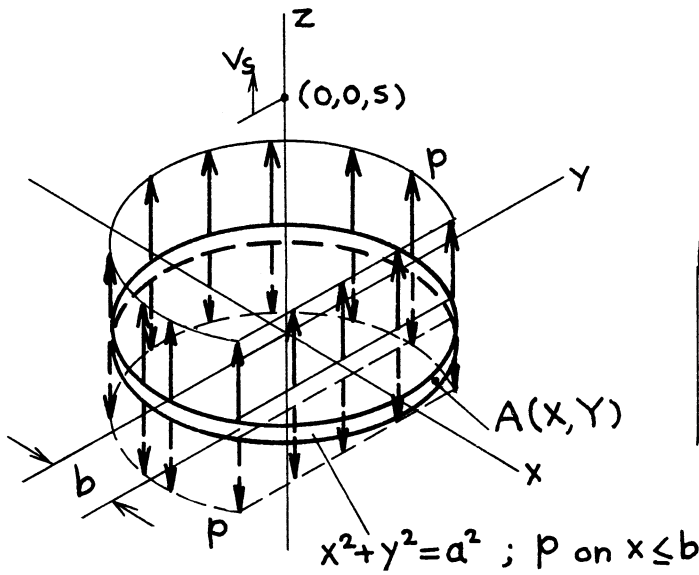
Volume of Crack:

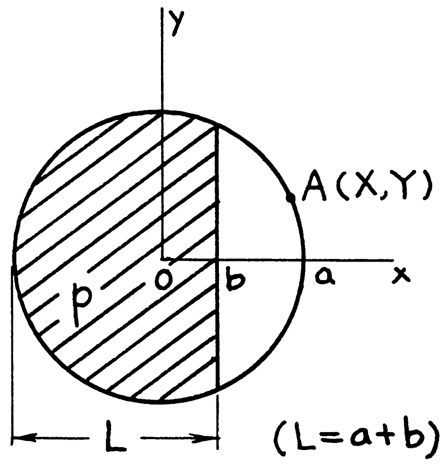
$$V = \frac{4\left(1-\nu^2\right)}{E} \sigma_0 \int_{-a}^{a} \left(a^2 - x^2\right) f(x) \, dx$$

Crack Opening at Center:

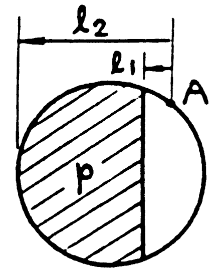
$$\delta_0 = 2v(0,0,0) = \frac{4\left(1-\nu^2\right)}{\pi E} \sigma_0 \int_{-a}^{a} f(x) \ell n \frac{a}{|x|} dx$$

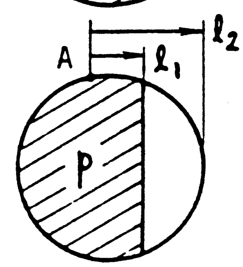
Method: Integration of **page 24.11**
Accuracy: Exact
References: **Tada 1975, 1985**





$$K_{IA} = \begin{cases} \dfrac{p}{\sqrt{\pi a}}\left\{\sqrt{\ell_2} - \sqrt{\ell_1}\right\}^2 & \\ \left(= \dfrac{p}{\sqrt{\pi a}}\left\{\sqrt{a+X} - \sqrt{X-b}\right\}^2\right) & X > b \\ \dfrac{p}{\sqrt{\pi a}}\left\{2a - \left(\sqrt{\ell_2} - \sqrt{\ell_1}\right)^2\right\} & \\ \left(= \dfrac{p}{\sqrt{\pi a}}\left\{2a - \left(\sqrt{a-X} - \sqrt{b-X}\right)^2\right\}\right) & X < b \end{cases}$$





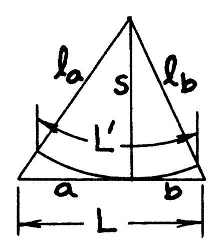
Volume of Crack:

$$V = \frac{4\left(1-\nu^2\right)}{3E} pL^2 (3a - L)$$

Crack Opening at Center:

$$\delta_0 = 2v(0,0,0) = \frac{4\left(1-\nu^2\right)}{\pi E} p\left\{L - b\,\ell n\,\frac{a}{|b|}\right\}$$

Vertical Displacement at $(0, 0, s)$:



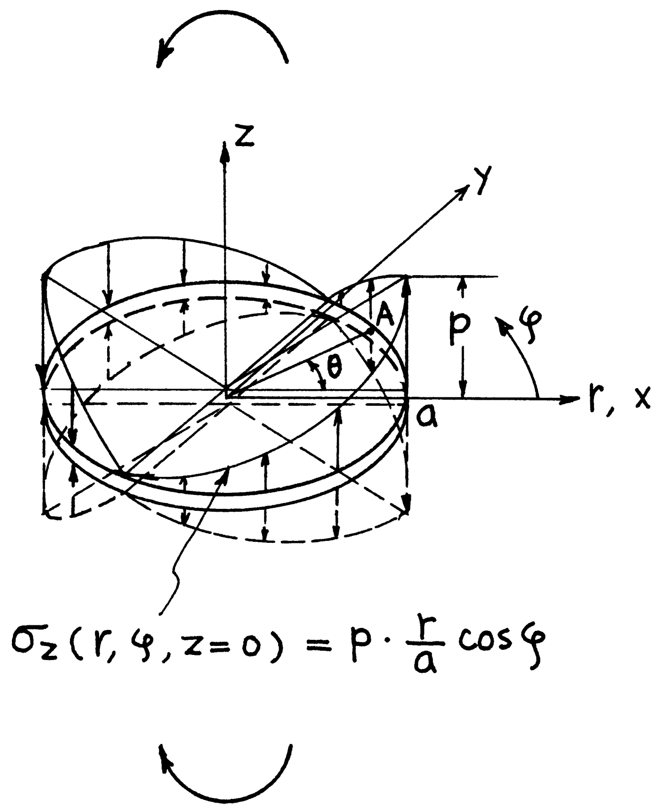
$$v_0 = v(0,0,s) = \frac{2\left(1-\nu^2\right)}{\pi E} p\left(1 - \alpha s\frac{\partial}{\partial s}\right)\left\{b\,\ell n\,\frac{\ell_a}{\ell_b} + L - L'\right\}$$

$$= \frac{2\left(1-\nu^2\right)}{\pi E} p\left(1 - \alpha s\frac{\partial}{\partial s}\right)\left\{b\,\ell n\sqrt{\frac{a^2+s^2}{b^2+s^2}} + (a+b) - s\left(\tan^{-1}\frac{a}{s} + \tan^{-1}\frac{b}{s}\right)\right\}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

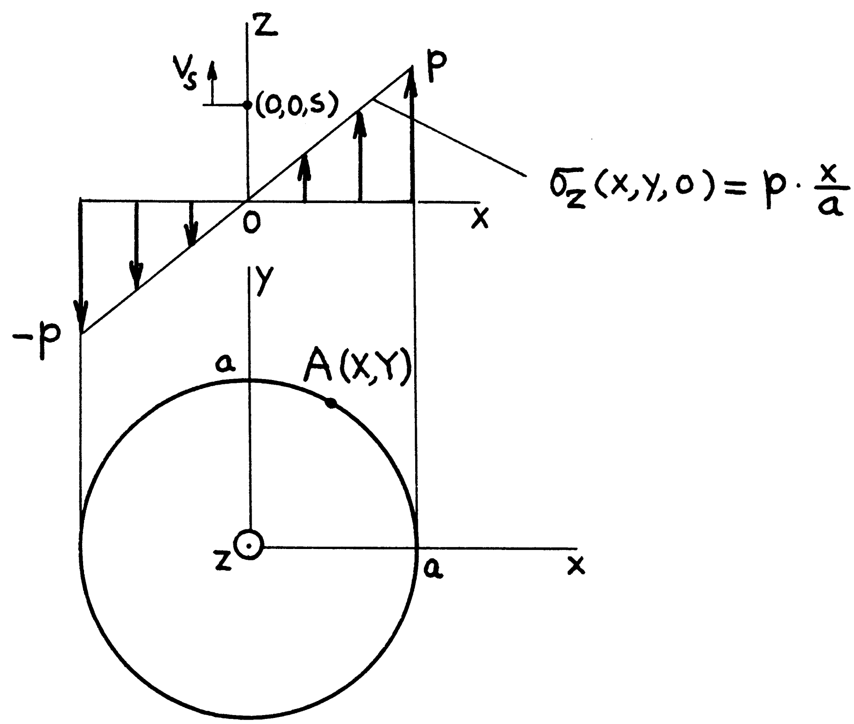
Method: Integration of **page 24.11**
Accuracy: Exact
Reference: **Tada 1985**



$$K_{IA} = \frac{4}{3\pi} p \sqrt{\pi a} \quad \cos\theta$$

$$K_{II} = K_{III} = 0$$

Method: Integral Transform (Hankel Transform) or Integration of **page 24.9 or 24.11**
Accuracy: Exact
Reference: **Benthem 1972**



$$K_{IA} = \frac{4}{3\pi} p \sqrt{\pi a} \cdot \frac{X}{a}$$

Volume of Crack:

$$V = 0$$

Crack Opening along Diameter $x = 0$:

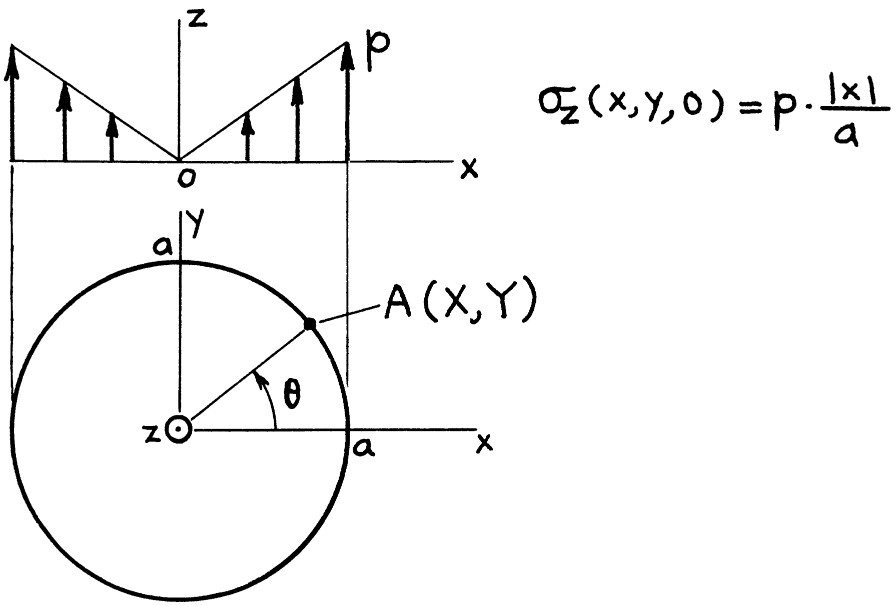
$$\delta = 2v(0, y, 0) = 0$$

Vertical Displacement at $(0, 0, s)$:

$$v_s = v(0, 0, s) = 0$$

Method: Integration of **page 24.11**
Accuracy: Exact
Reference: **Tada 1985**

NOTE: Crack surface interference $(x < 0)$ was not considered.



$$K_{IA} = \frac{2}{\pi} p\sqrt{\pi a}\left\{\frac{1}{6} + \frac{4}{3}\left(\frac{X}{a}\right)^{3/2}\left(\sqrt{1+\frac{X}{a}} - \sqrt{\frac{X}{a}}\right)\right\}$$

or

$$= \frac{2}{\pi} p\sqrt{\pi a}\left\{\frac{1}{6} + \frac{4}{3}(\cos\theta)^{3/2}\left(\sqrt{1+\cos\theta} - \sqrt{\cos\theta}\right)\right\}$$

Volume of Crack:

$$V = \frac{2\,(1-\nu^2)}{E} pa^3 \left[= \frac{16\,(1-\nu^2)}{3E} pa^3 \cdot \left(\frac{3}{8}\right)\right]$$

Crack Opening at Center:

$$\delta_0 = 2v\,(0,0,0) = \frac{2\left(1-\nu^2\right)}{\pi E}\, pa\left[= \frac{8\left(1-\nu^2\right)}{\pi E}\, pa \cdot \left(\frac{1}{4}\right)\right]$$

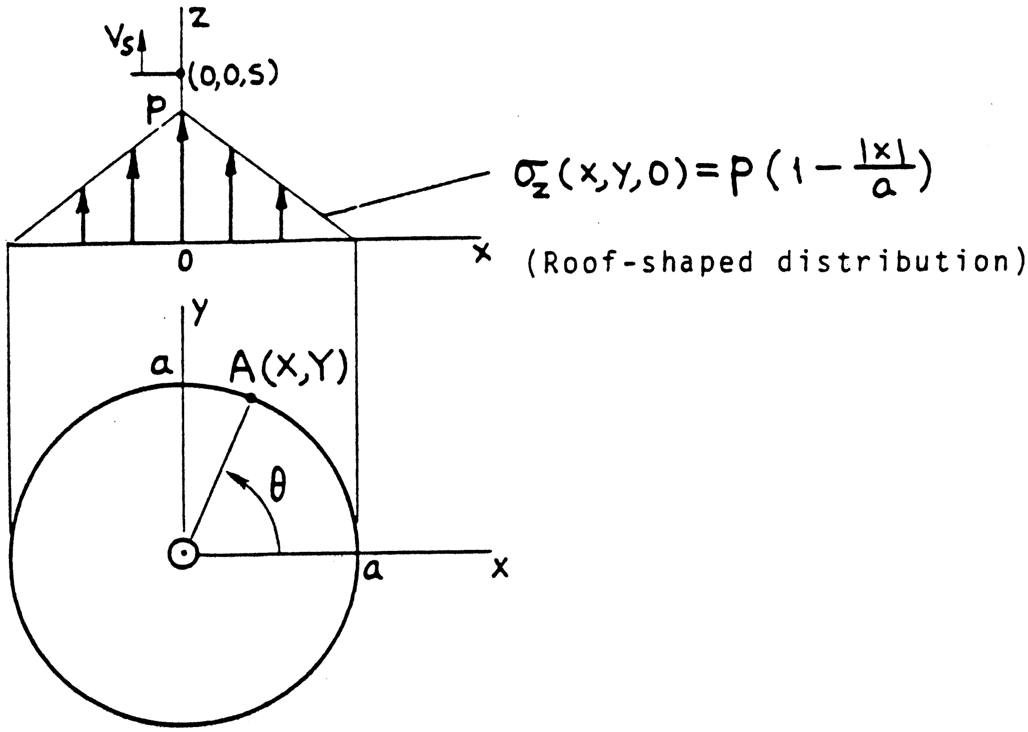
Displacement at $(0,0,s)$:

$$v_s = v\,(0,0,s) = \frac{1-\nu^2}{\pi E}\, pa\left\{1 - (1-2\,\alpha)\frac{s^2}{a^2}\,\ell n\left(1+\frac{a^2}{s^2}\right) - 2\,\alpha\frac{s^2}{s^2+a^2}\right\}$$

where

$$\alpha = \frac{1}{2\,(1-\nu)}$$

Method: Integration of **page 24.11**
Accuracy: Exact
Reference: **Tada 1975, 1985**



$$K_{IA} = \frac{2}{\pi} p\sqrt{\pi a}\left\{\frac{5}{6} - \frac{4}{3}\left(\frac{X}{a}\right)^{3/2}\left(\sqrt{1+\frac{X}{a}} - \sqrt{\frac{X}{a}}\right)\right\}$$

or

$$= \frac{2}{\pi} p\sqrt{\pi a}\left\{\frac{5}{6} - \frac{4}{3}(\cos\theta)^{3/2}\left(\sqrt{1+\cos\theta} - \sqrt{\cos\theta}\right)\right\}$$

Volume of Crack:

$$V = \frac{10\left(1-\nu^2\right)}{3E} pa^3 \left[= \frac{16\left(1-\nu^2\right)}{3E} pa^3 \cdot \left(\frac{5}{8}\right)\right]$$

Crack Opening at Center:

$$\delta_0 = 2v(0,0,0) = \frac{6\left(1-\nu^2\right)}{\pi E} pa \left[= \frac{8\left(1-\nu^2\right)}{\pi E} pa \cdot \left(\frac{3}{4}\right)\right]$$

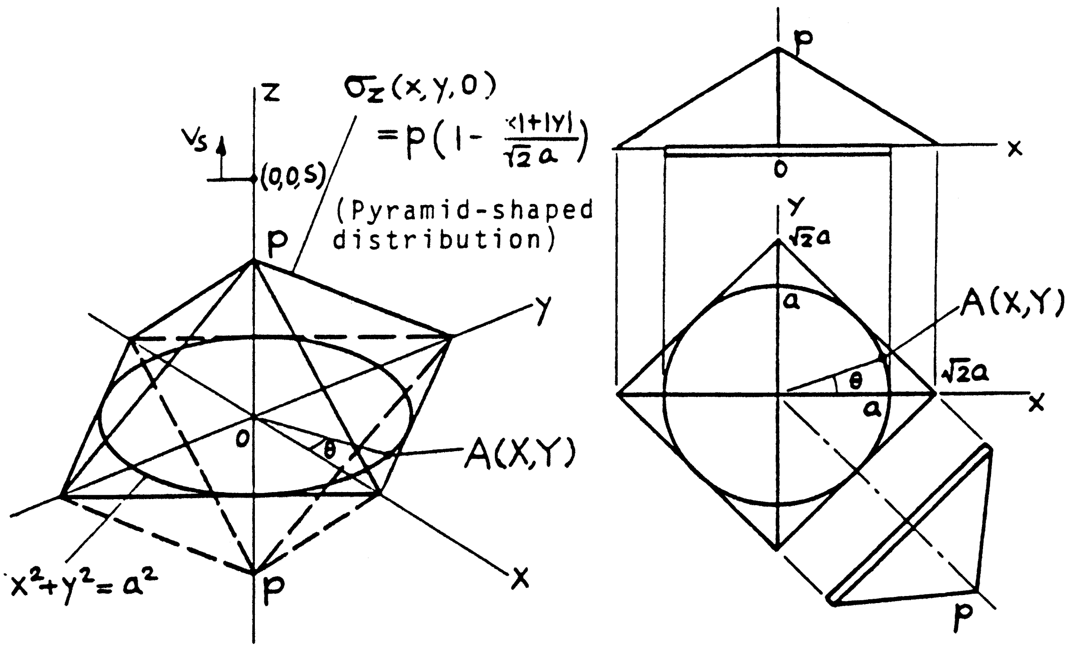
Displacement at $(0,0,s)$:

$$v_s = v(0,0,s) = \frac{1-\nu^2}{\pi E} pa \left\{3 - 4(1-\alpha)\frac{s}{a}\tan^{-1}\frac{a}{s} - (1-2\alpha)\frac{s^2}{a^2}\ell n\left(1+\frac{a^2}{s^2}\right) - 2\alpha\frac{s^2}{s^2+a^2}\right\}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Method: Superposition of **pages 24.1 and 24.15**.
Accuracy: Exact
References: **Tada 1975, 1985**



$$K_{IA} = \frac{2}{\pi} p\sqrt{\pi a}\left\{\left(1+\frac{1}{\sqrt{2}}\right) - \frac{2\sqrt{2}}{3}\left[\left(\frac{|X|}{a}\right)^{3/2}\sqrt{1+\frac{|X|}{a}} + \left(\frac{|Y|}{a}\right)^{3/2}\sqrt{1+\frac{|Y|}{a}}\right]\right\}$$

$$= \frac{2}{\pi} p\sqrt{\pi a}\left\{\left(1+\frac{1}{\sqrt{2}}\right) - \frac{2\sqrt{2}}{3}\left[(|\cos\theta|)^{3/2}\sqrt{1+|\cos\theta|} + (|\sin\theta|)^{3/2}\sqrt{1+|\sin\theta|}\right]\right\}$$

Volume of Crack:

$$V = \frac{16\left(1-\nu^2\right)}{3E} pa^3 \cdot \left(1 - \frac{3}{8}\sqrt{2}\right)\left[= \frac{16\left(1-\nu^2\right)}{3E} pa^3 \cdot (.4697)\right]$$

Crack Opening at Center:

$$\delta_0 = \frac{8\left(1-\nu^2\right)}{\pi E} pa \cdot \left(1 - \frac{\sqrt{2}}{4}\right)\left[= \frac{8\left(1-\nu^2\right)}{\pi E} pa \cdot (.6464)\right]$$

Displacement at $(0, 0, s)$:

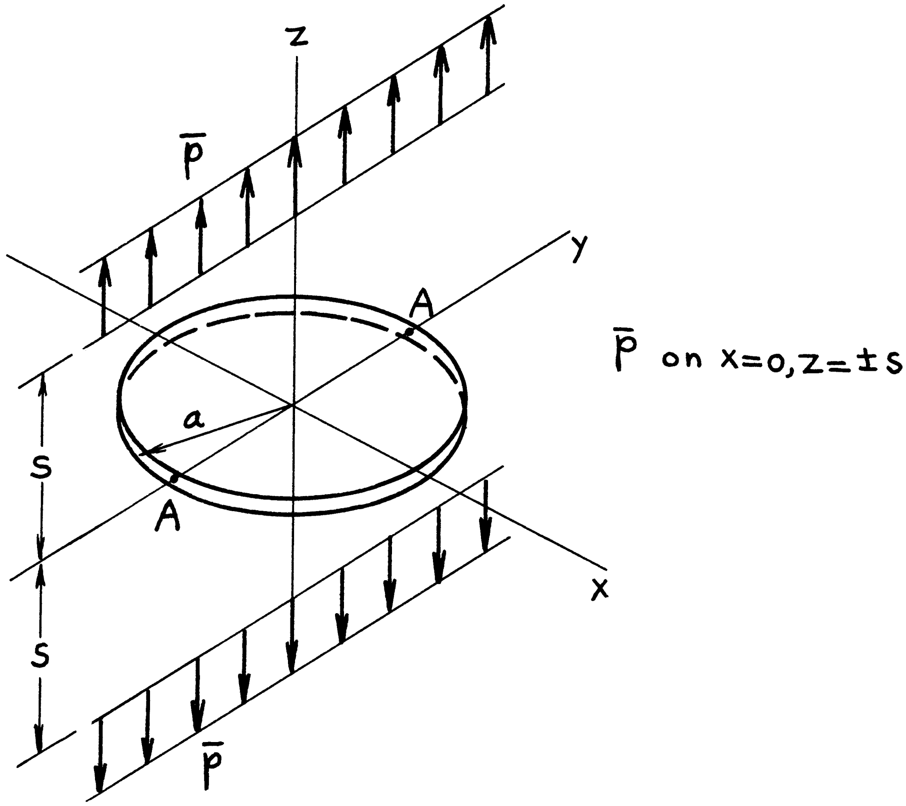
$$v_s = v(0,0,s)$$
$$= \frac{1-\nu^2}{\pi E} pa\left\{\left(4-\sqrt{2}\right) - 4(1-\alpha)\frac{s}{a}\tan^{-1}\frac{a}{s} + \sqrt{2}(1-2\alpha)\frac{s^2}{a^2}\ell n\left(1+\frac{a^2}{s^2}\right) - 2\left(2-\sqrt{2}\right)\alpha\frac{s^2}{s^2+a^2}\right\}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Method: Superposition of **page 24.16**
Accuracy: Exact
Reference: **Tada 1985**

NOTE: Because of symmetry, it is only necessary to calculate K_{IA} for $0 \leq \theta \leq \pi/4$.



Maximum Value of K_I:

$$(K_I)_{\max} = K_{IA} = \frac{\sqrt{2}\,\overline{p}}{(\pi)^{3/2}}\left(1 - \alpha s\frac{\partial}{\partial s}\right)\left\{\frac{1}{\sqrt{s}}\left(\tanh^{-1}\frac{\sqrt{2as}}{s+a} + \tan^{-1}\frac{\sqrt{2as}}{s-a}\right) - \frac{\sqrt{2}}{\sqrt{a}}\tan^{-1}\frac{a}{s}\right\}$$

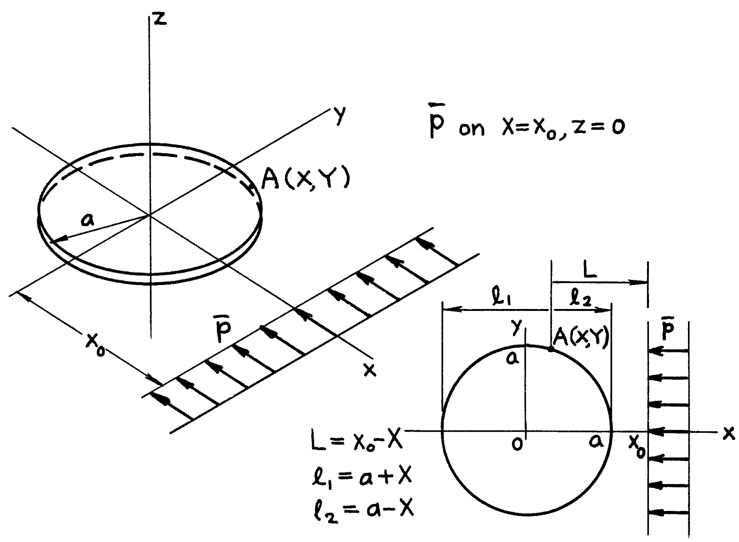
Volume of Crack:

$$V = \frac{8\left(1-\nu^2\right)}{\pi E}\overline{p}\,a^2\left\{\left[1 + (1-2\alpha)\frac{s^2}{a^2}\right]\tan^{-1}\frac{a}{s} - (1-2\,\alpha)\frac{s}{a}\right\}$$

where

$$\alpha = \frac{1}{2\,(1-\nu)}$$

Method: Integration of **page 24.11**
Accuracy: Exact
Reference: **Tada 1985**



$$K_{IA} = \frac{\overline{p}\,(1-\alpha)}{\pi\sqrt{\pi a}}\left\{\sqrt{\frac{\ell_1}{L}}\tan^{-1}\sqrt{\frac{\ell_1}{L}} + \sqrt{\frac{\ell_2}{L}}\tanh^{-1}\sqrt{\frac{\ell_2}{L}} - \tanh^{-1}\frac{a}{x_0}\right\}$$

$$(K_{II} = K_{III} = 0)$$

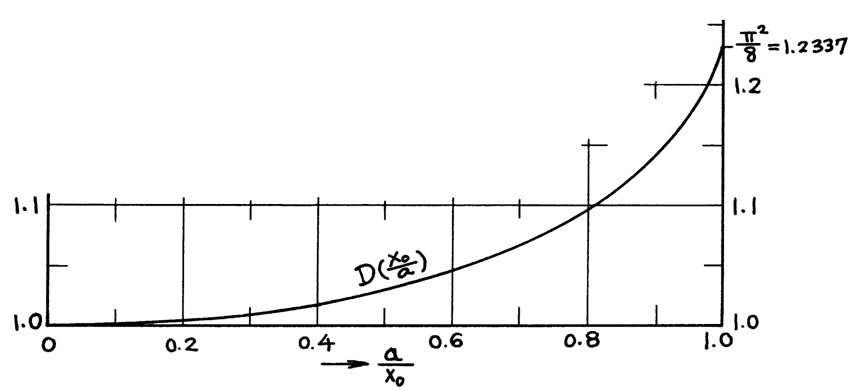
Volume of Crack:

$$V = \frac{4\left(1-\nu^2\right)}{\pi E}\,\overline{p}\,(1-\alpha)a^2\left\{\frac{x_0}{a} - \left[\left(\frac{x_0}{a}\right)^2 - 1\right]\tanh^{-1}\frac{a}{x_0}\right\}$$

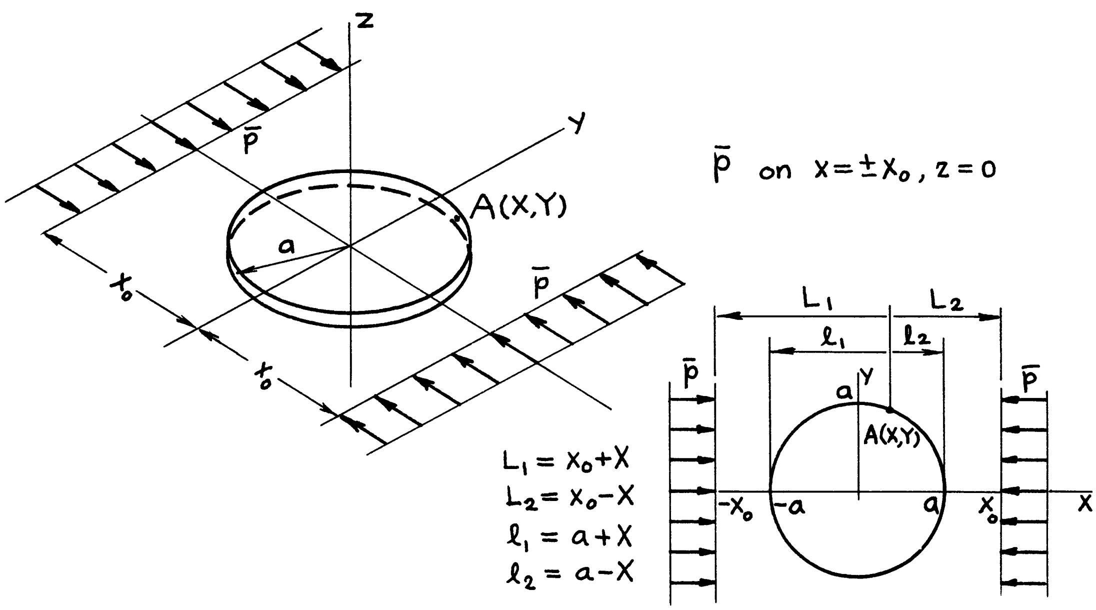
Crack Opening at Center:

$$\delta_0 = 2v\,(0,0,0) = \frac{8\left(1-\nu^2\right)}{\pi E}\left(\frac{\overline{p}(1-\alpha)}{2\pi}\cdot\frac{1}{x_0}\right)a\cdot D\left(\frac{x_0}{a}\right)$$

where $\alpha = \dfrac{1}{2\,(1-\nu)}$



Method: Integration of **page 24.11**
Accuracy: K_{IA}, V Exact; $D(^{x_0}/_a)$ curve is based on accurate numerical values.
Reference: **Tada 1985**



$$K_{IA} = \frac{\overline{p}(1-\alpha)}{\pi\sqrt{\pi a}} \left\{ \left(\sqrt{\frac{\ell_1}{L_1}} \tanh^{-1} \sqrt{\frac{\ell_1}{L_1}} + \sqrt{\frac{\ell_2}{L_2}} \tanh^{-1} \sqrt{\frac{\ell_2}{L_2}} \right) \right.$$
$$\left. + \left(\sqrt{\frac{\ell_2}{L_1}} \tan^{-1} \sqrt{\frac{\ell_2}{L_1}} + \sqrt{\frac{\ell_1}{L_2}} \tan^{-1} \sqrt{\frac{\ell_1}{L_2}} \right) - 2\tanh^{-1} \frac{a}{x_0} \right\}$$

$$(K_{II} = K_{III} = 0)$$

Volume of Crack:

$$V = \frac{8\left(1-\nu^2\right)}{\pi E} \overline{p}(1-\alpha)a^2 \left\{ \frac{x_0}{a} - \left[\left(\frac{x_0}{a} \right)^2 - 1 \right] \tanh^{-1} \frac{a}{x_0} \right\}$$

Crack Opening at Center:

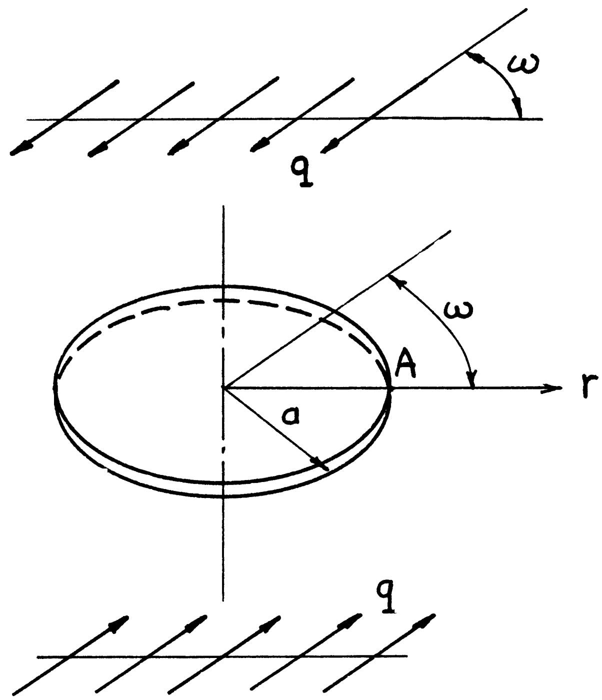
$$\delta_0 = 2v(0,0,0) = \frac{8\left(1-\nu^2\right)}{\pi E} \left(\frac{\overline{p}(1-\alpha)}{\pi} \cdot \frac{1}{x_0} \right) a \cdot D\left(\frac{x_0}{a} \right)$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Method: Superposition of **page 24.19**
Accuracy: K_{IA}, V Exact; $D(x_0/a)$ curve is based on accurate numerical values.
Reference: **Tada 1985**

NOTE: For numerical values of $D(x_0/a)$, see **page 24.19**.

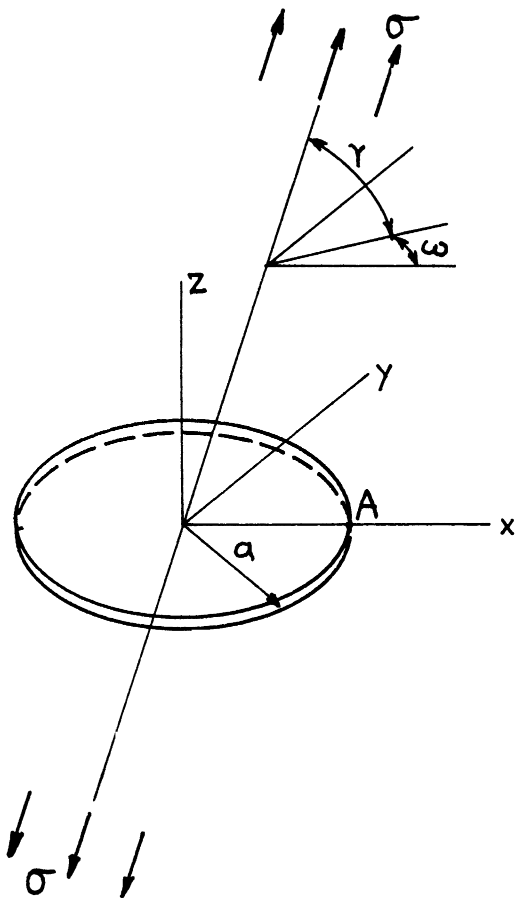


$$K_{IA} = 0$$

$$K_{IIA} = \frac{4}{\pi(2-\nu)}(q\cos\omega)\sqrt{\pi a}$$

$$K_{IIIA} = \frac{4(1-\nu)}{\pi(2-\nu)}(q\sin\omega)\sqrt{\pi a}$$

Method: Three-Dimensional Potential Functions or a Special Case of Elliptical Crack (**page 26.3**)
Accuracy: Exact
References: **Segedin 1950; Sih 1968**

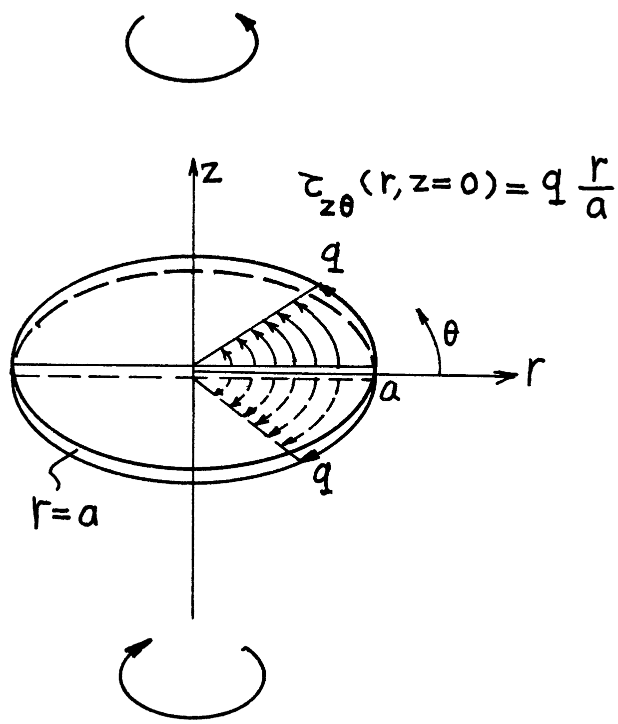


$$K_{IA} = \frac{2}{\pi}\left(\sigma \sin^2 \gamma\right)\sqrt{\pi a}$$

$$K_{IIA} = \frac{4}{\pi(2-\nu)}(\sigma \sin\gamma \cos\gamma)\cos\omega\sqrt{\pi a}$$

$$K_{IIIA} = \frac{4(1-\nu)}{\pi(2-\nu)}(\sigma \sin\gamma \cos\gamma)\sin\omega\sqrt{\pi a}$$

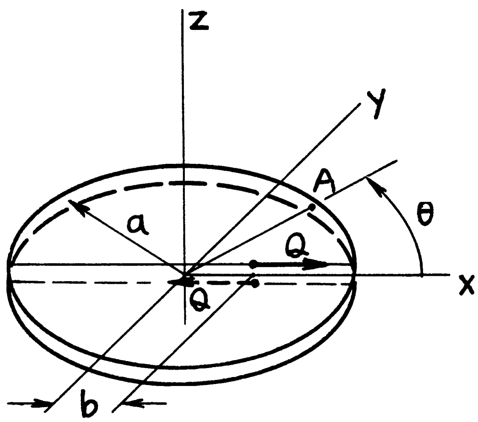
Method: Superposition of **pages 24.1 and 24.21** or Special Case of Elliptical Crack (**page 26.2**)
Accuracy: Exact
References (for Elliptical Crack): **Sadowski 1949; Green 1950; Irwin 1962b; Sih 1968**

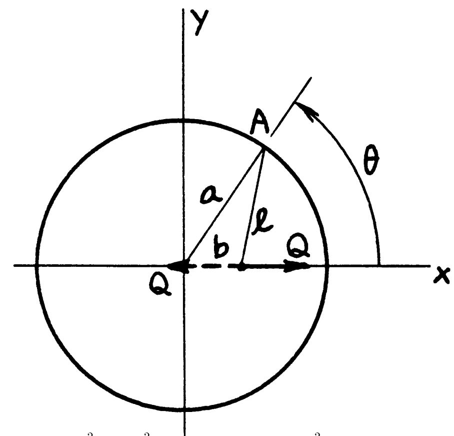


$$K_I = K_{II} = 0$$

$$K_{III} = \frac{4}{3\pi} q\sqrt{\pi a}$$

Method: Integral Transform (Hankel Transform) or from Stress Concentration Factor
Accuracy: Exact
References: **Neuber 1937; Weinstein 1952; Collins 1962; Benthem 1972**





$\ell^2 = a^2 - 2ab\cos\theta + b^2$

$$B = {}^{b}\!/_{a}$$

$$L = {}^{\ell}\!/_{a}\left(L^2 = 1 - 2B\cos\theta + B^2\right)$$

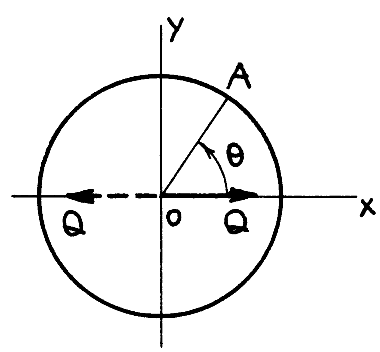
$$K_I = 0$$

$$K_{IIA} = \frac{Q}{(\pi a)^{3/2}} \cdot \frac{1}{\sqrt{1-B^2}} \cdot \left\{ B + \frac{2}{2-\nu} \cdot \frac{1}{L^2}\left[\left\{1 + (1-\nu)B^2\right\}(\cos\theta - B) + \nu\left(1-B^2\right)\frac{\left(1+B^2\right)\cos\theta - 2B}{L^2}\right]\right\}$$

$$K_{IIIA} = -\frac{2Q}{(\pi a)^{3/2}} \cdot \frac{\sqrt{1-B^2}}{2-\nu} \cdot \frac{\sin\theta}{L^2}\left\{1 - \nu - \nu\frac{1-B^2}{L^2}\right\}$$

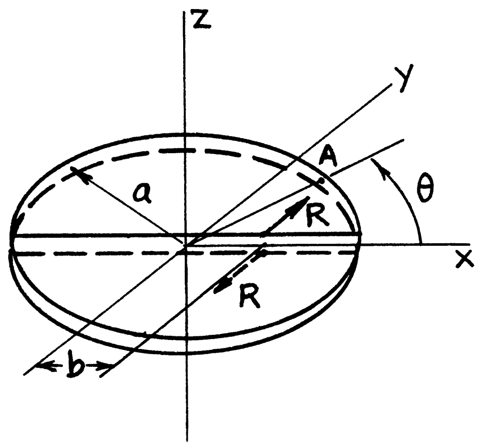
$\underline{b=0}$ $$K_{IIA} = \frac{2Q}{(\pi a)^{3/2}} \cdot \frac{1+\nu}{2-\nu}\cos\theta$$

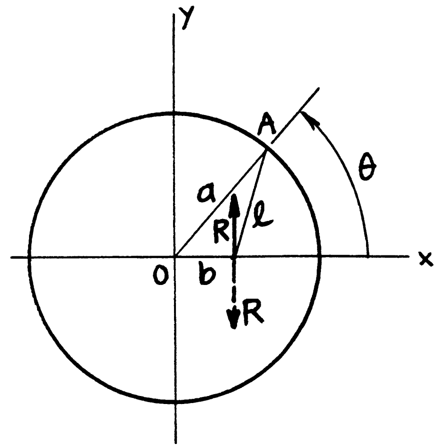
$$K_{IIIA} = -\frac{2Q}{(\pi a)^{3/2}} \cdot \frac{1-2\nu}{2-\nu}\sin\theta$$



(See **page 24.25**)
Method: Fourier Series Expansions
Accuracy: Exact
References: **Kassir 1975; Tada 1985**

NOTE: A minor error in K_{III} (**Kassir 1975**) was corrected.





$$\ell^2 = a^2 - 2ab\cos\theta + b^2$$

$$B = {}^{b}\!/_{a}$$

$$L = {}^{\ell}\!/_{a}\left(L^2 = 1 - 2B\cos\theta + B^2\right)$$

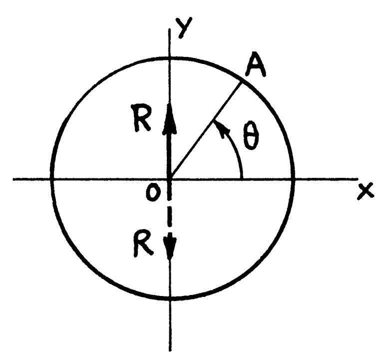
$$K_I = 0$$

$$K_{IIA} = \frac{2R}{(\pi a)^{3/2}} \cdot \frac{1}{2-\nu} \cdot \frac{1+B^2}{\sqrt{1-B^2}} \cdot \frac{\sin\theta}{L^2}\left\{1 + \nu\frac{1-B^2}{L^2}\right\}$$

$$K_{IIIA} = \frac{2R}{(\pi a)^{3/2}} \cdot \frac{1}{2-\nu} \cdot \frac{1}{\sqrt{1-B^2}} \cdot \frac{1}{L^2}\left\{\left(1-\nu+B^2\right)\left(\cos\theta - B\right) - \nu\left(1-B^2\right)\frac{\left(1+B^2\right)\cos\theta - 2B}{L^2}\right\}$$

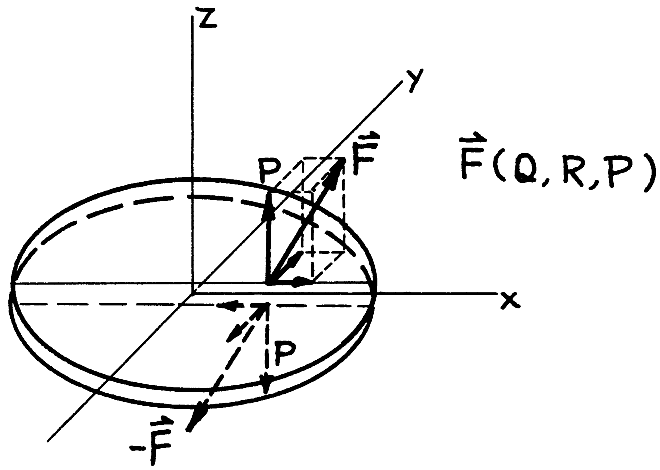
$\underline{b=0}$ $$K_{IIA} = \frac{2R}{(\pi a)^{3/2}} \cdot \frac{1+\nu}{2-\nu}\sin\theta$$

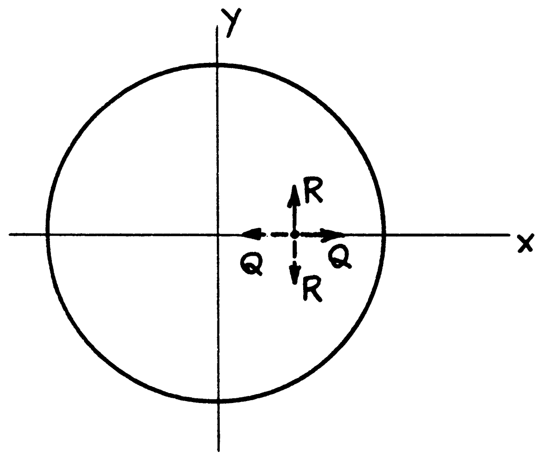
$$K_{IIIA} = \frac{2R}{(\pi a)^{3/2}} \cdot \frac{1-2\nu}{2-\nu}\cos\theta$$



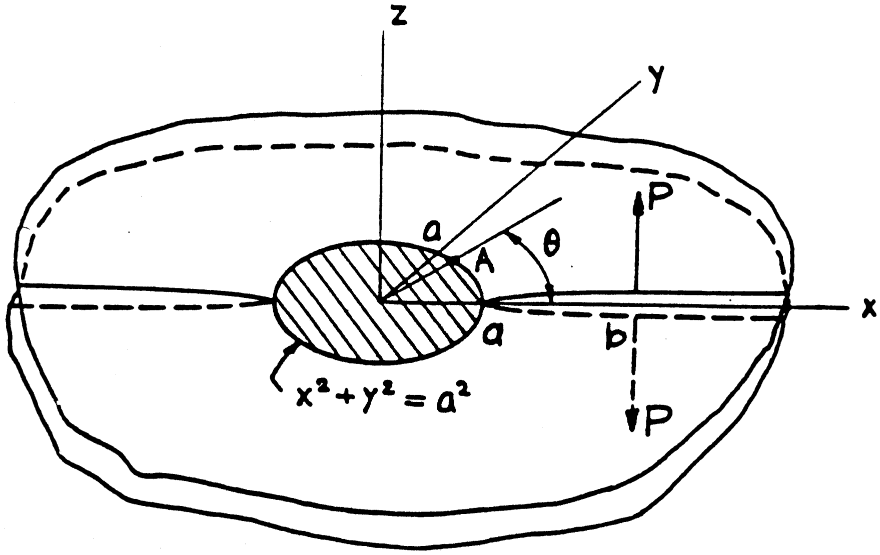
(See **page 24.24**)
Method: Fourier Series Expansions
Accuracy: Exact
References: **Kassir 1975; Tada 1985**

NOTE: The series form solutions in **Kassir (1975)** were converted into closed-form expressions.





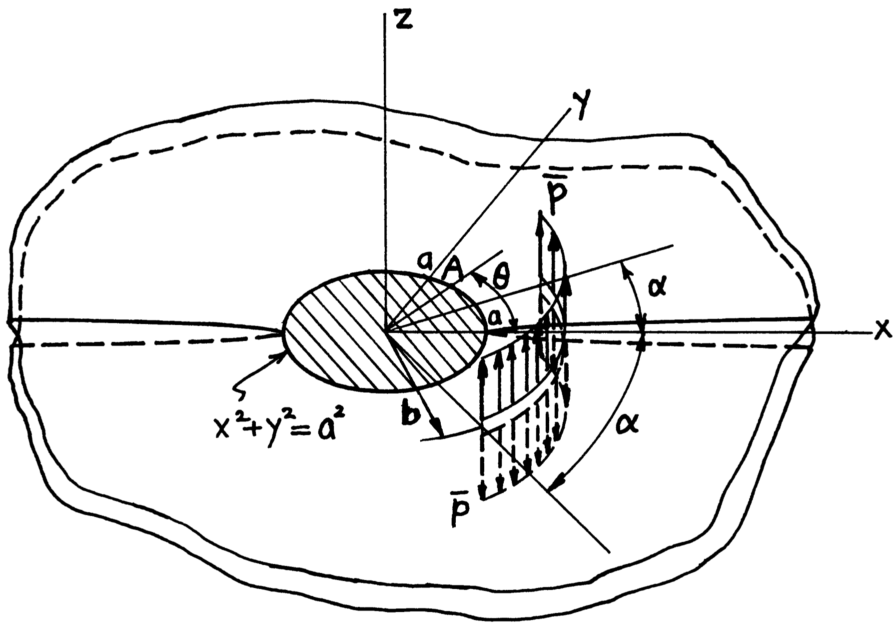
See **pages 24.3, 24.24, and 24.25**.



$$K_{IA} = \frac{P}{(\pi a)^{3/2}} \left\{ \cos^{-1}\frac{a}{b} + \frac{a\sqrt{b^2 - a^2}}{a^2 + b^2 - 2ab\cos\theta} \right\}$$

$$K_{IIA} = K_{IIIA} = 0$$

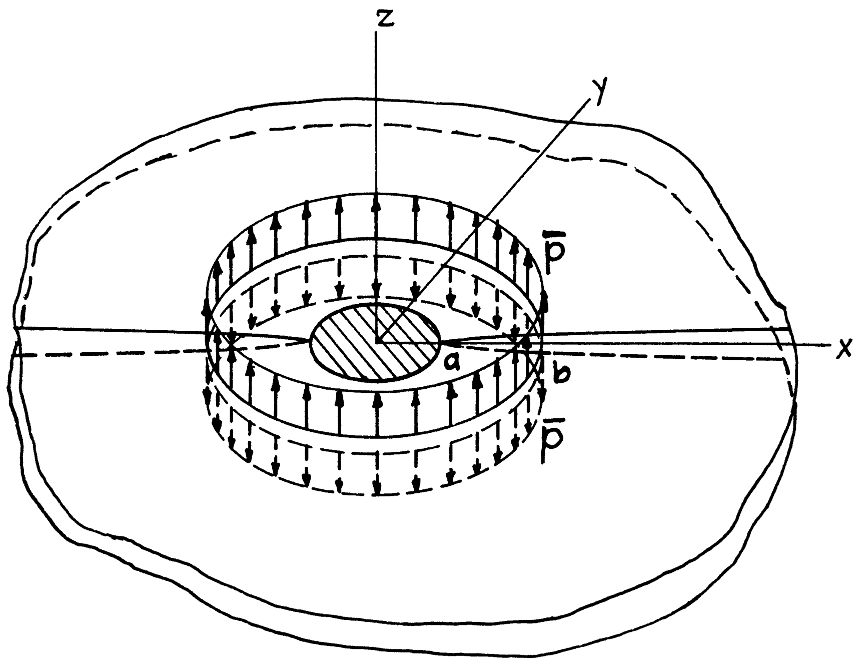
Method: Neuber-Papkovich potential
Accuracy: Exact
References: **Galin 1953; Tada 1973**



$$K_{IA} = \frac{2\bar{p}b}{(\pi a)^{3/2}}\left[\alpha \cos^{-1}\frac{a}{b} + \frac{a}{\sqrt{b^2-a^2}}\left\{\tan^{-1}\left(\frac{b+a}{b-a}\tan\frac{\theta+\alpha}{2}\right) - \tan^{-1}\left(\frac{b+a}{b-a}\tan\frac{\theta-\alpha}{2}\right)\right\}\right]$$

$$K_{IIA} = K_{IIIA} = 0$$

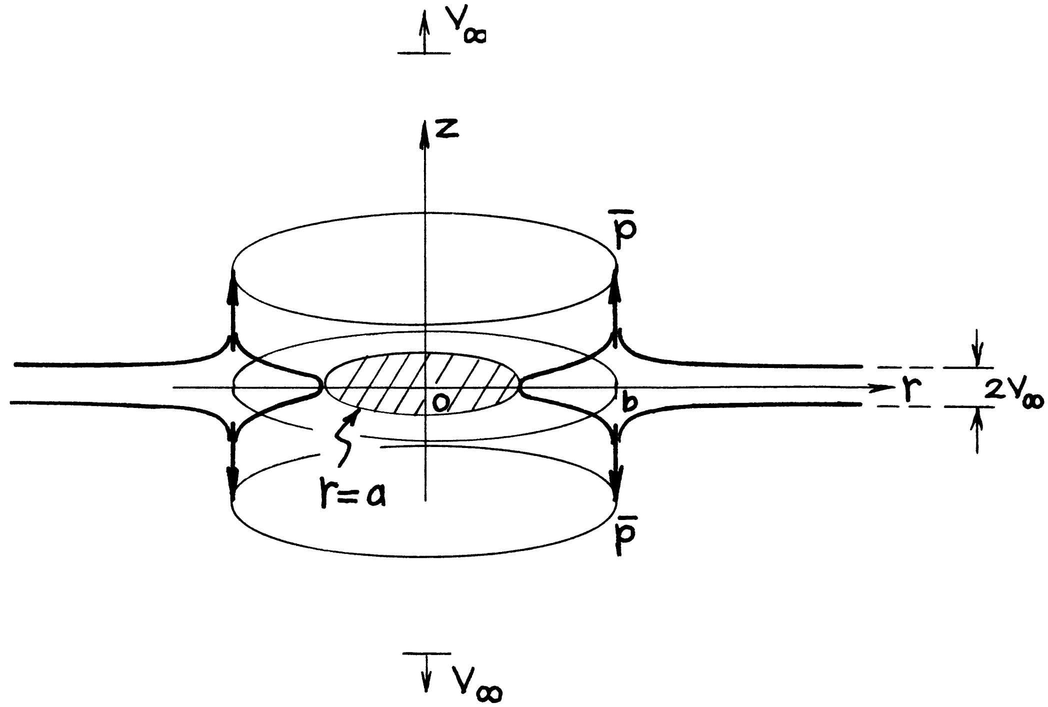
Method: Integration of **page 25.1**
Accuracy: Exact
Reference: **Tada 1973**



$$K_I = \frac{\overline{p}(2\pi b)}{(\pi a)^{3/2}} \left\{ \cos^{-1}\frac{a}{b} + \frac{a}{\sqrt{b^2 - a^2}} \right\}$$

$$K_{II} = K_{III} = 0$$

Methods: Boussinesq-Papkovich Potential or Special Case of **page 25.2**
Accuracy: Exact
References: **Bueckner 1972; Tada 1973**

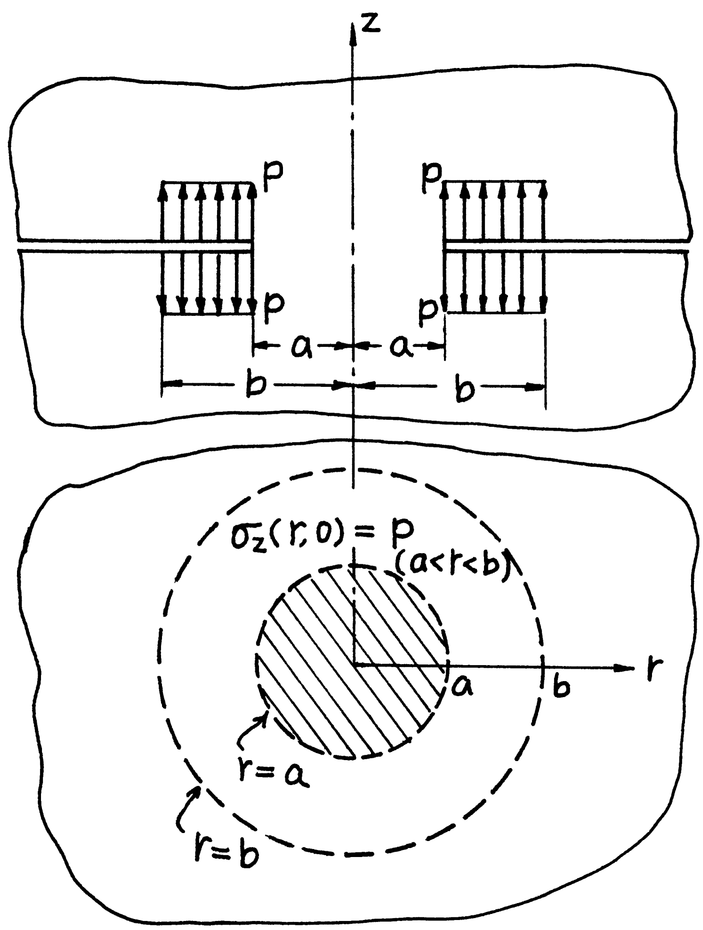


$$K_I = \frac{2\overline{p}}{\sqrt{\pi a}}\left\{\frac{b}{a}\cos^{-1}\frac{a}{b} + \frac{b}{\sqrt{b^2 - a^2}}\right\}$$

Relative Displacement at Infinity:

$$2V_\infty = \frac{4\left(1-\nu^2\right)}{E}\,\overline{p}\cdot\frac{b}{a}\cos^{-1}\frac{a}{b}$$

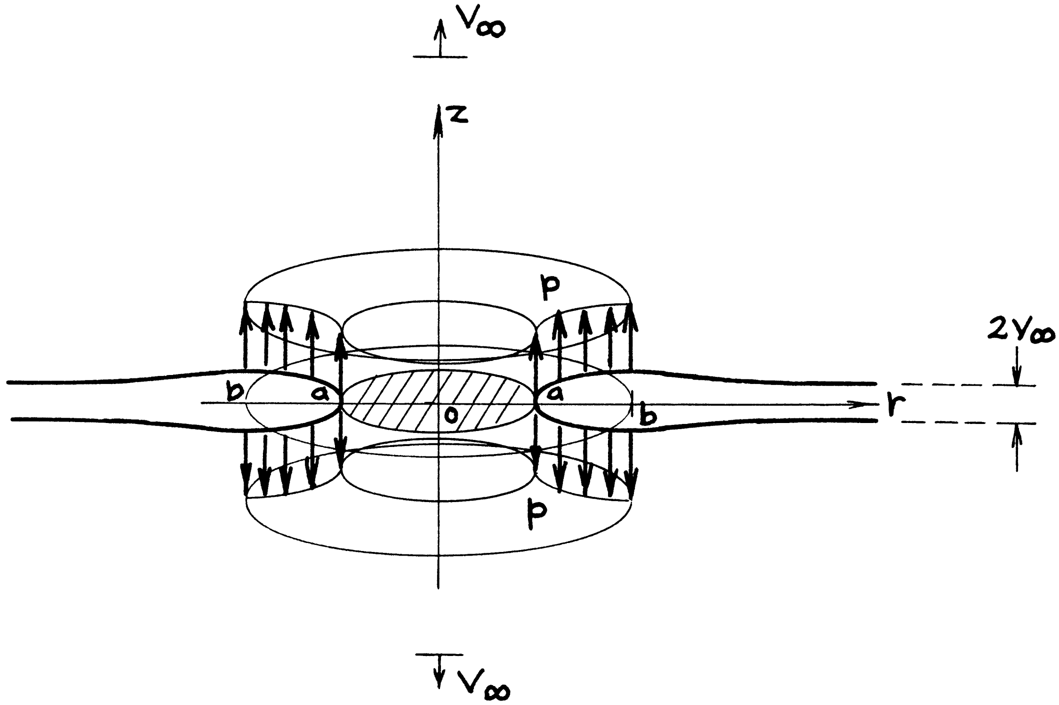
Method: Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**



$$K_I = \frac{P\left(\pi b^2\right)}{(\pi a)^{3/2}}\left\{\cos^{-1}\frac{a}{b} + \frac{a}{b}\sqrt{1-\left(\frac{a}{b}\right)^2}\right\}$$

$$K_{II} = K_{III} = 0$$

Method: Integration of **page 25.3**
Accuracy: Exact
Reference: **Tada 1973**

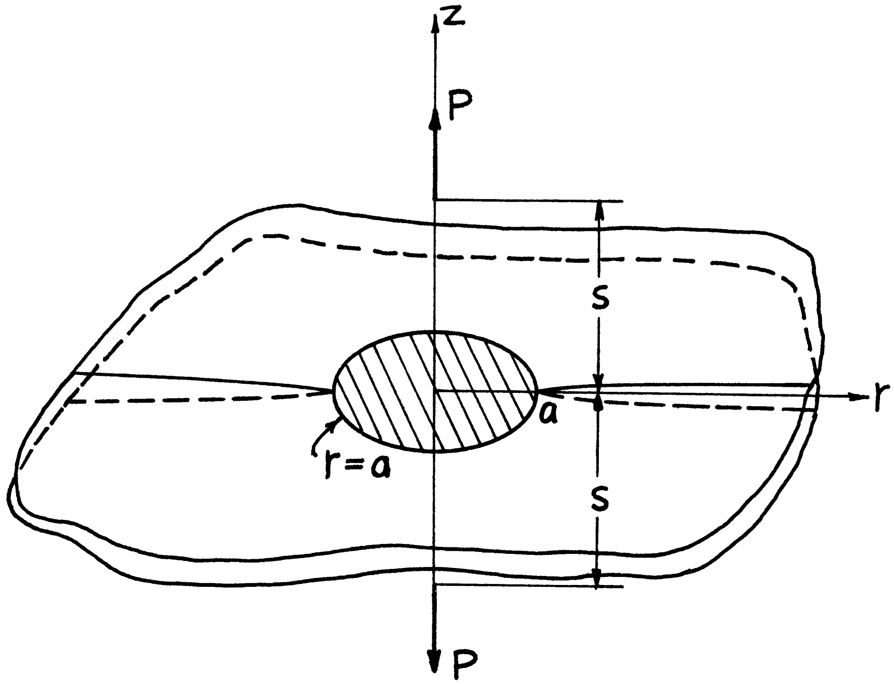


$$K_I = \frac{P}{\sqrt{\pi}}\sqrt{a}\left\{\left(\frac{b}{a}\right)^2 \cos^{-1}\frac{a}{b} + \sqrt{\left(\frac{b}{a}\right)^2 - 1}\right\}$$

Relative Displacement at Infinity:

$$2V_\infty = \frac{2(1-\nu^2)}{E}\,pa\left\{\left(\frac{b}{a}\right)^2 \cos^{-1}\frac{a}{b} - \sqrt{\left(\frac{b}{a}\right)^2 - 1}\right\}$$

Method: Integration of **page 25.3** or Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**



$$\underset{No\ Crack}{\sigma_z (r, z = 0)} = \frac{P}{2\pi}\left[1 - \alpha s \frac{\partial}{\partial s}\right] \frac{s}{\left(r^2 + s^2\right)^{3/2}}$$

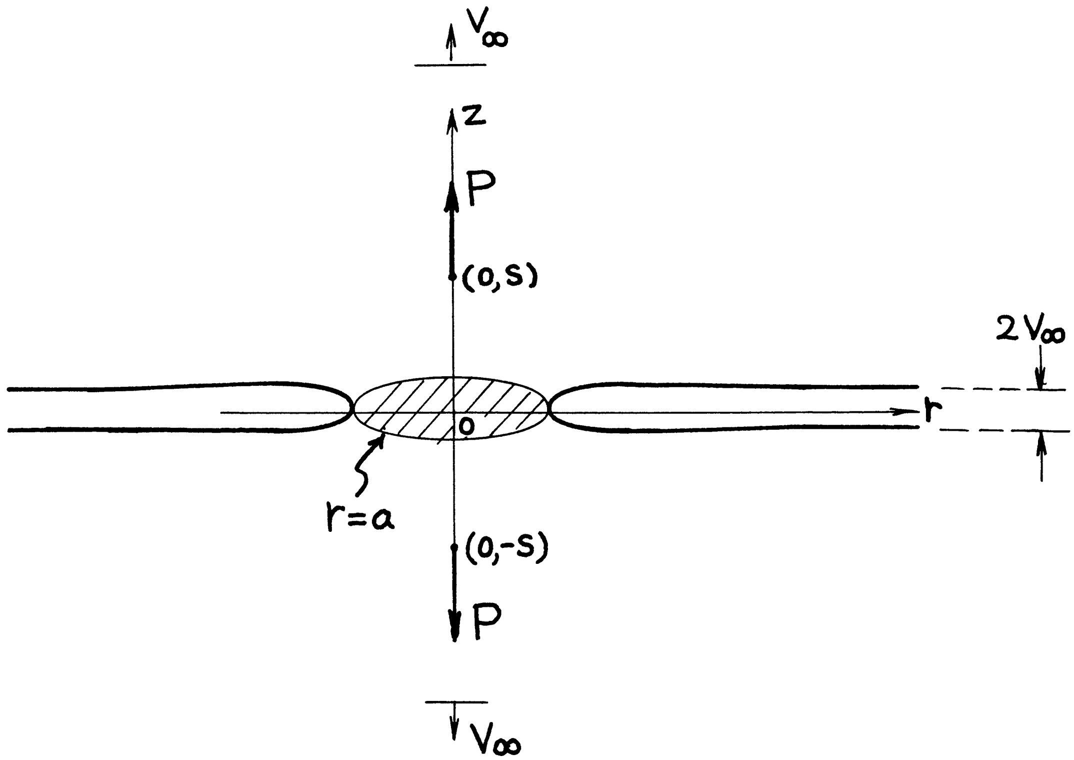
$$K_I = \frac{P}{(\pi a)^{3/2}}\left[\tan^{-1}\frac{s}{a} + \frac{as}{a^2 + s^2}\left\{1 - \alpha \frac{2a^2}{a^2 + s^2}\right\}\right]$$

$$K_{II} = K_{III} = 0$$

where

$$\alpha = \frac{1}{2(1 - \nu)}$$

Method: Integration of **page 25.3**
Accuracy: Exact
Reference: **Tada 1973**



$$K_I = \frac{P}{(\pi a)^{3/2}}\left(1 - \alpha s\frac{\partial}{\partial s}\right)\left\{\tan^{-1}\frac{s}{a} + \frac{as}{a^2+s^2}\right\}$$

$$or \qquad = \frac{P}{(\pi a)^{3/2}}\left\{\tan^{-1}\frac{s}{a} + \frac{s/a}{1+(s/a)^2} - \alpha\frac{s/a}{\left[1+(s/a)^2\right]^2}\right\}$$

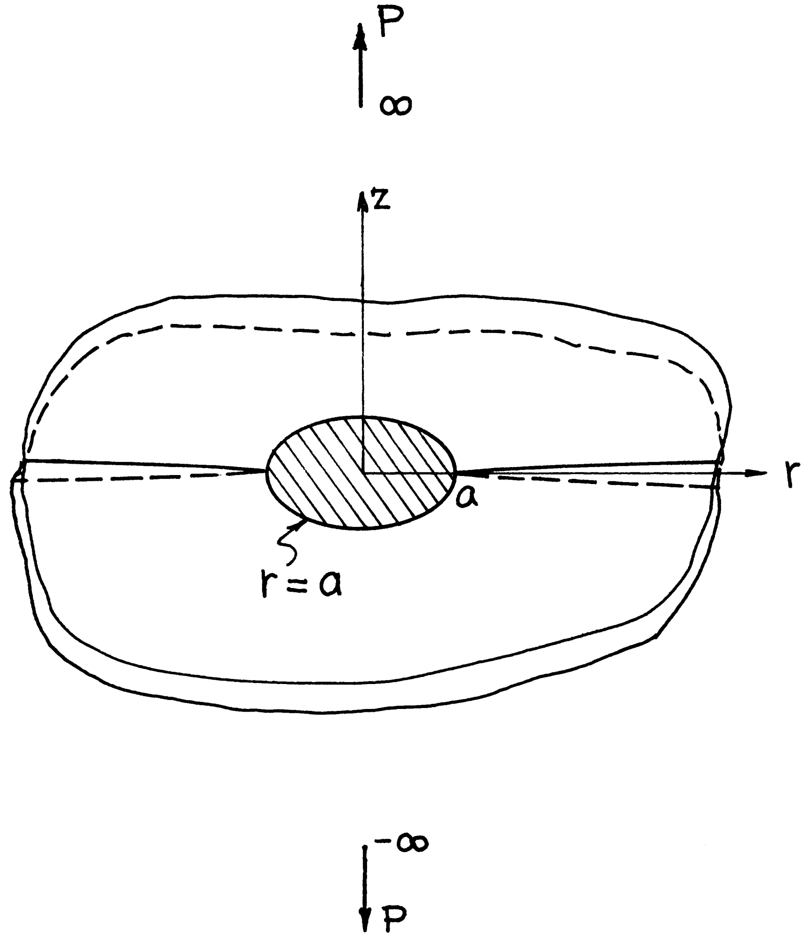
Relative Displacement at Infinity:

$$2V_\infty = \frac{2\left(1-\nu^2\right)}{\pi E}\frac{P}{a}\left\{\tan^{-1}\frac{s}{a} - \alpha\frac{s/a}{1+(s/a)^2}\right\}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Method: Reciprocity (see **page 25.6a**) or Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**



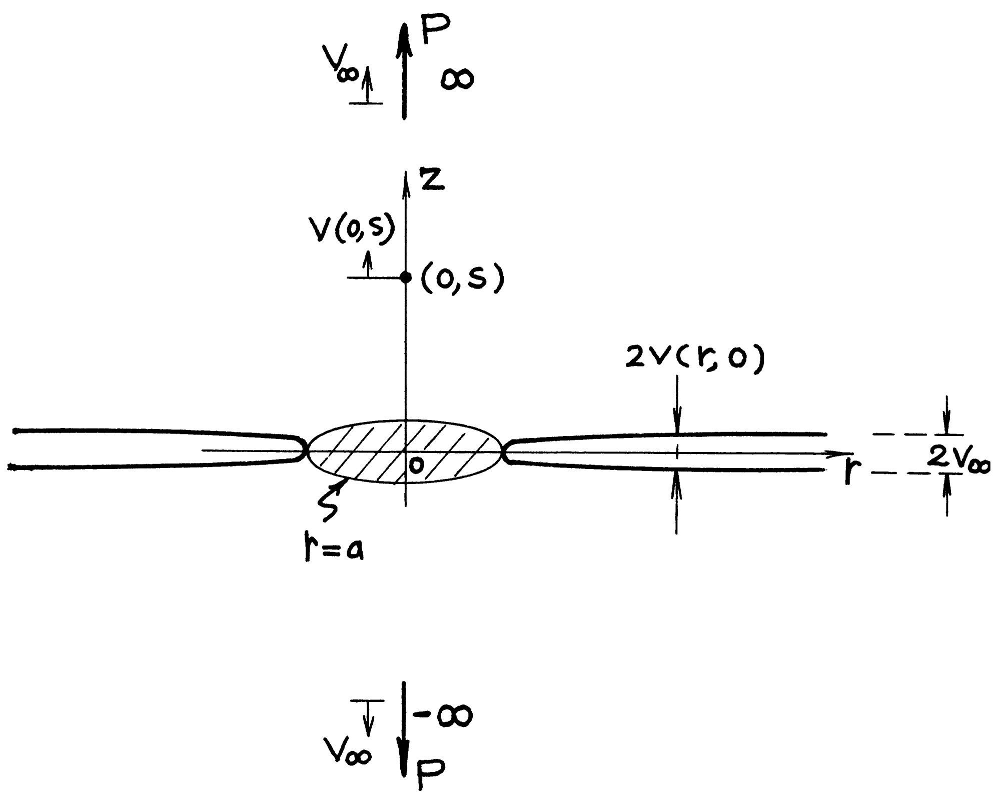
$$K_I = \frac{P}{2a\sqrt{\pi a}}$$

$$K_{II} = K_{III} = 0$$

Methods: Stress Concentration Factor (Neuber), Solution for a Stamp Problem (e.g., Sneddon), Special Case of Elliptical Net Ligament (**page 26.4**) or Special Case of **page 25.5**

Accuracy: Exact

References: **Neuber 1937; Sneddon 1951**



$$K_I = \frac{P}{2a\sqrt{\pi a}}$$

Crack Opening Profile:

$$\underset{r \geq a}{2v\ (r, 0)} = \frac{2\left(1-\nu^2\right)}{\pi E}\frac{P}{a}\cos^{-1}\frac{a}{r}$$

Vertical Displacement at $(0, s)$:

$$v(0, s) = \frac{1-\nu^2}{\pi E}\frac{P}{a}\left\{\tan^{-1}\frac{s}{a} - \alpha\frac{s/a}{1+(s/a)^2}\right\}$$

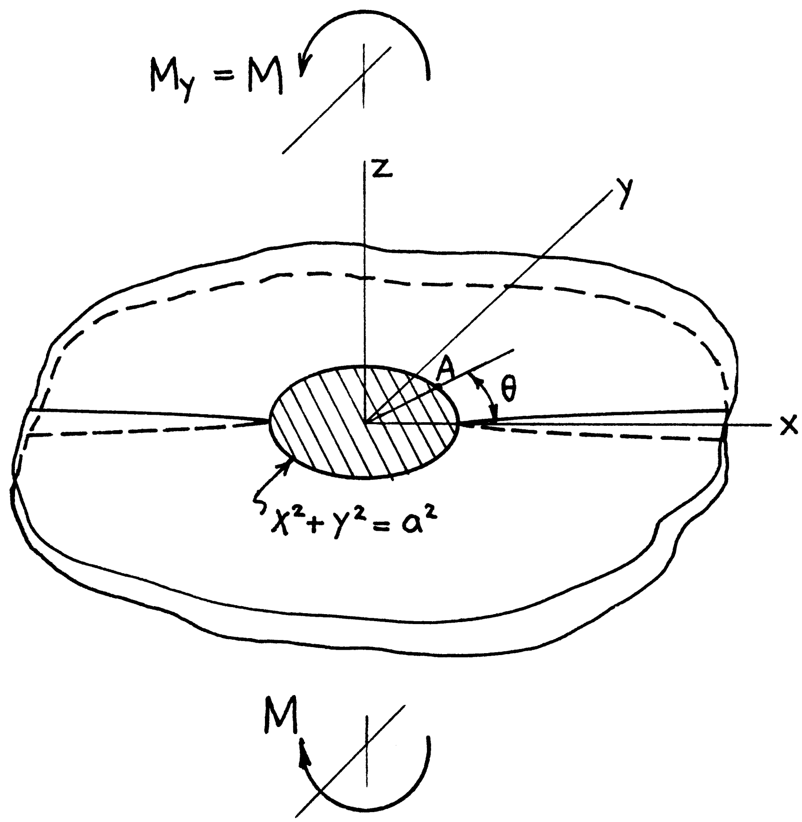
Relative Displacement at Infinity:

$$2V_\infty\left(= \underset{r\to\infty}{\{2v(r, 0)\}} = \underset{s\to\infty}{\{2v(0, s)\}}\right) = \frac{1-\nu^2}{E}\frac{P}{a}$$

where

$$\alpha = \frac{1}{2(1-\nu)}$$

Method: Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**

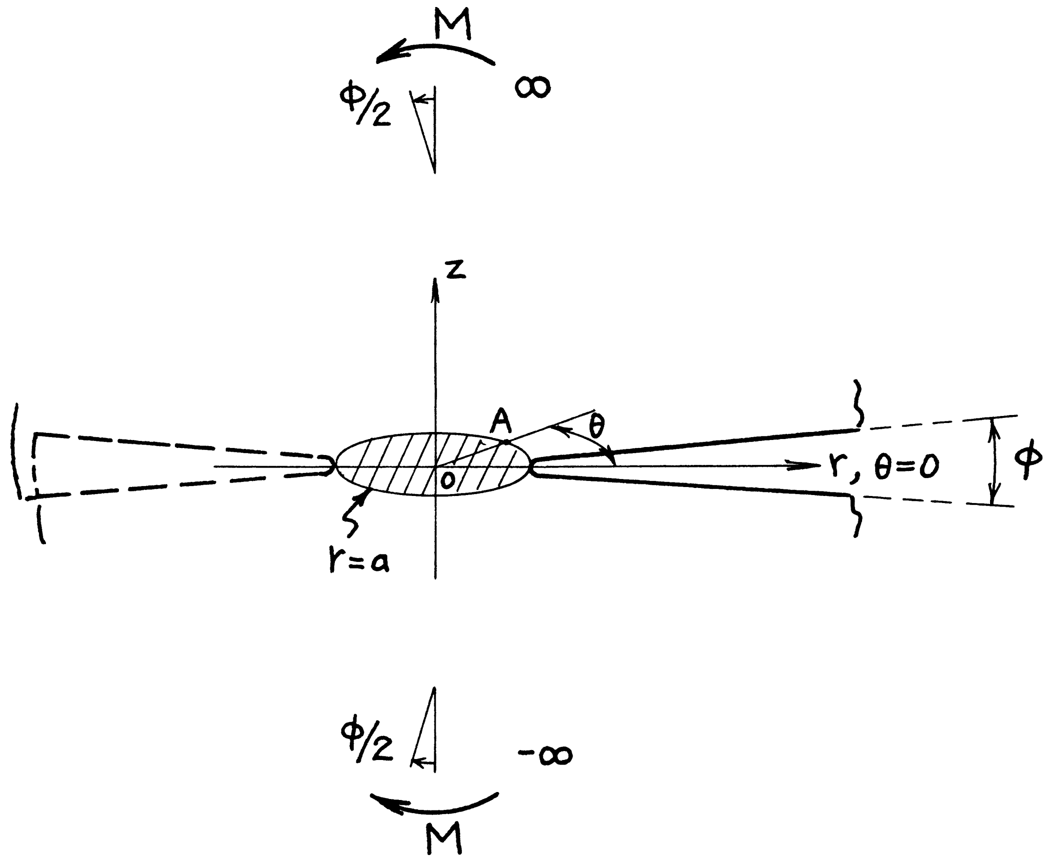


$$K_{IA} = \frac{3}{2}\frac{M}{a^2\sqrt{\pi a}}\cos\theta$$

$$= \frac{3}{8}\sigma_0\sqrt{\pi a}\cos\theta \qquad \left(\sigma_0 = \frac{4M}{\pi a^3}\right)$$

$$K_{II} = K_{III} = 0$$

Methods: Stress Concentration Factor (Neuber), Solution for a Stamp Problem (e.g., Sneddon)
Accuracy: Exact
References: **Neuber 1937; Sneddon 1951**



$$K_{IA} = \frac{3}{2}\frac{M}{a^2\sqrt{\pi a}}\cos\theta$$

Relative Rotation at Infinity:

$$\phi = \frac{3(1-\nu^2)}{2E}\cdot\frac{M}{a^3}$$

K_{IA} in terms of ϕ:

$$K_{IA} = \frac{E}{\sqrt{\pi}\left(1-\nu^2\right)}\phi\sqrt{a}\cos\theta$$

Method: Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**

NOTE: Crack surface interference ($\pi/2 < \theta < 3\pi/2$) was not considered.